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Topological Quantum Computing

Exploring Majorana Zero Modes or something subtitley

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subtitley

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Abstract

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Chapter 1

Introduction

Quantum computing offers the possibility of solving classes of problems that are intractable on classical hardware, but this promise is limited by one central difficulty: quantum states are fragile. Decoherence, noise, and imperfect control make it challenging to preserve and manipulate quantum information long enough to perform useful computations. One of the most attractive proposed solutions to this problem is topological quantum computing, where information is stored non-locally in collective quantum states that are intrinsically less sensitive to local perturbations. In this picture, protection is not achieved only through better engineering, but through the structure of the quantum states themselves.

Majorana zero modes have therefore attracted broad interest as candidate building blocks for fault-tolerant quantum computing. In condensed-matter systems, they appear not as fundamental particles, but as emergent zero-energy quasiparticles in topological superconductors [2]. Their most important feature is that a pair of spatially separated Majorana modes can encode a non-local fermionic degree of freedom. Because local perturbations cannot easily access this information, such states offer a route toward more robust quantum operations. In addition, Majorana modes are predicted to obey non-Abelian exchange statistics, meaning that exchanging them can implement quantum gates through braiding rather than through purely local control pulses.

Although the basic theoretical picture is well established, realizing and controlling Majorana modes in realistic devices remains a major challenge. Much of the experimental and theoretical effort has therefore focused on engineered hybrid systems, where conventional superconductivity, tunneling, spin structure, and confinement can be combined to produce effective topological behavior [1]. Among the simplest of these platforms is the quantum-dot–superconductor–quantum-dot setup, often referred to as the Poor Man's Majorana system. This device can be understood as a minimal analogue of the Kitaev chain: hopping between dots plays the role of the kinetic term, non-local pairing induced by the superconductor plays the role of effective p-wave pairing, and gate-controlled dot energies act as local chemical potentials.

The appeal of this platform is twofold. First, it is conceptually simple enough to connect directly to the Kitaev-chain picture, which makes it well suited for theoretical analysis. Second, it is experimentally tunable: dot energies, effective pairing strengths, and tunnel couplings can all be adjusted in a controlled way. At the same time, this simplicity comes with limitations. In finite and interacting systems, one should be careful not to overstate the meaning of low-energy degeneracies or Majorana-like observables. The relevant question is therefore not only whether a model displays signatures associated with Majorana physics, but whether those signatures survive under interactions and whether they remain useful for braiding operations.

This thesis studies that question in interacting chains of coupled quantum dots. More specifically, I investigate whether an interacting n-dot Poor Man's Majorana chain can be numerically optimized to produce low-energy states with the expected Majorana-like characteristics, and whether those optimized states support a useful braiding protocol. The goal is not to claim exact topological protection in a finite microscopic model. Rather, the aim is to identify parameter regimes where the low-energy structure is sufficiently close to the Majorana ideal to make the physical interpretation meaningful and the braiding behavior nontrivial.

To do this, I model the system with an interacting quantum-dot Hamiltonian containing on-site energies, nearest-neighbor hopping, induced pairing, and Coulomb repulsion. The search for promising parameter regimes is formulated as an optimization problem, where the loss function is built from several physically motivated Majorana criteria: near-degeneracy between parity sectors, finite low-energy gaps, small charge differences between parity partners, and a Majorana-polarization profile consistent with edge-localized modes. Once optimized configurations are found, I analyze their parity-resolved spectra and local diagnostics, reconstruct effective Majorana operators from the low-energy manifold, and compare an ideal four-Majorana braiding protocol with a braid generated by projected microscopic junction operators.

The main results show a clear distinction between weakly and strongly interacting regimes. The optimization reproduces the known non-interacting two-dot sweet spot and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. Among the interacting cases studied here, the three-dot system at weak interaction emerges as the most convincing candidate: it combines a very small ground-state splitting, a clear gap to excited states, and a spatial Majorana profile concentrated at the chain edges. In the lowest projected manifold, the corresponding microscopic braid agrees remarkably well with the ideal effective-model braid. However, this agreement deteriorates as stronger interactions and higher-energy sectors are included, indicating that useful braiding depends sensitively on how well the exchange remains confined to a clean low-energy subspace.

The broader motivation of the thesis is therefore not only to identify candidate Majorana-like configurations, but also to clarify the boundary between effective low-energy success and microscopic breakdown. In that sense, the work is positioned between abstract Majorana models and realistic device-level questions: the optimization tells us which interacting parameter regimes are promising, while the braiding analysis tests whether those regimes support the operation that ultimately matters for topological quantum computing.

The remainder of the thesis is organized as follows. Chapter 2 presents the theoretical background, beginning with superconductivity and the Bogoliubov-de Gennes formalism, then introducing the Kitaev chain, the quantum-dot realization of Poor Man's Majoranas, and the basic logic of braiding. Chapter 3 describes the numerical methods used to optimize the interacting dot chains, analyze the resulting low-energy states, and construct the braiding protocol. Chapter 4 presents the optimization results, local Majorana diagnostics, and braiding benchmarks in both the idealized and projected microscopic models. Chapter 5 discusses the physical interpretation of these findings, their limitations, and the role of interactions in shaping the low-energy manifold. Finally, Chapter 6 summarizes the main conclusions and outlines possible directions for future work.

Chapter 2

Theory

2.1 Majorana Zero Modes and Topological Motivation

Topological quantum computing is an approach to fault-tolerant quantum computation in which quantum gates are implemented by braiding non-Abelian anyons. The central idea is that information can be encoded non-locally in a degenerate subspace, so that local perturbations have limited access to the encoded quantum state. In this setting, the result of a gate depends on the topology of the braid rather than on microscopic details of the path.
Anyons are quasiparticles that, so far, have only been observed in effectively two-dimensional systems. In three spatial dimensions, identical particles are classified as either bosons or fermions, corresponding to symmetric and antisymmetric exchange statistics. In two dimensions, however, the topology of particle exchange allows for a richer possibility: exchanging two particles can produce statistics that are neither bosonic nor fermionic
https://sciencex.com/wire-news/347971706/finally-anyons-reveal-their-exotic-quantum-properties.html
. In particular, exchanging two identical particles twice is not necessarily topologically equivalent to doing nothing, because the exchange path cannot always be continuously shrunk to a point without crossing another particle. Particles with such exchange statistics are called anyons, and they are classified as Abelian or non-Abelian depending on how the quantum state transforms under exchange. Abelian anyons acquire only a phase factor, while non-Abelian anyons transform the state within a degenerate subspace. This latter property is what makes non-Abelian anyons especially interesting for quantum computationhttps://www.nature.com/articles/nphys767.

A collection of non-Abelian anyons with fixed positions and fixed local quantum numbers can possess a non-trivial topological degeneracy. Quantum information can then be encoded in this degenerate subspace, while braiding the anyons implements unitary transformations within it. Since the resulting transformation depends only on the braid topology, and not on small local deformations of the path, this provides a possible route toward fault-tolerant quantum gates. In suitable schemes, braiding operations can be combined with additional ingredients to implement the gates needed for quantum computation.
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One of the most promising candidates for realizing non-Abelian exchange statistics

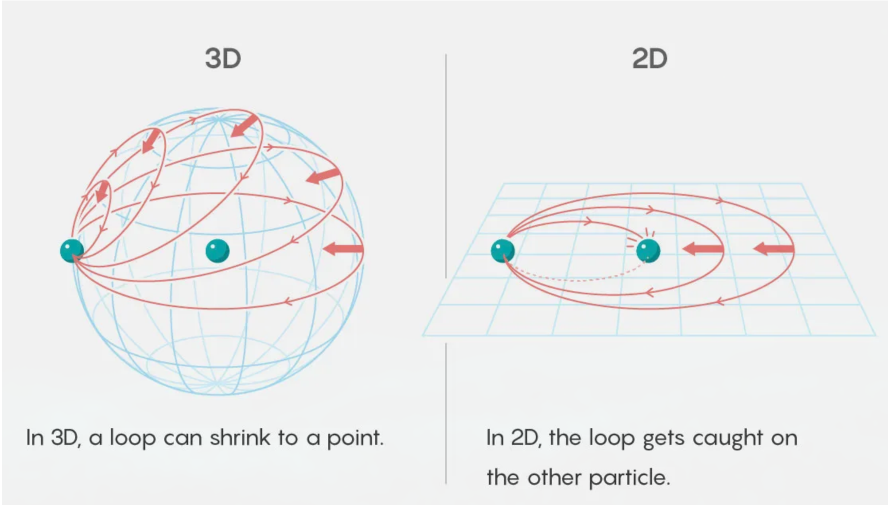

Figure 2.1: Visualization of the difference between particle exchange in two and three dimensions. In three dimensions, the exchange loop can be shrunk to a point, making a double exchange topologically equivalent to leaving the particles unchanged. In two dimensions, the corresponding loop cannot be shrunk to a point without crossing the particle worldlines. This topological distinction allows anyons to exhibit non-trivial exchange statistics.$https://media.wired.com/photos/5ebf0488d58e99fafced8087/master/w_1600\%2Cc_limit/Science_Anyons-graphic-FINAL.jpg$

in condensed-matter systems is the Majorana zero mode (MZM), predicted to occur in certain topological superconductors. A Majorana zero mode is a zero-energy quasiparticle described by an operator that is equal to its own Hermitian conjugate. In one-dimensional topological superconductors, such modes can appear at the ends of the system. Unlike an ordinary fermionic mode, a single MZM can be viewed as "half" of a Dirac fermion: two Majorana operators can be combined to form one conventional fermionic operator,

$$c = \frac{1}{2}(\gamma_1 + i\gamma_2) \tag{2.1}$$

with the inverse relations

$$\gamma_1 = c + c^\dagger, \qquad \gamma_2 = -i(c - c^\dagger). \tag{2.2}$$

Here γ_1 and γ_2 are Majorana operators satisfying

$$\gamma_i = \gamma_i^\dagger, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij} \tag{2.3}$$

The non-Abelian character of MZMs is reflected in the fact that exchanging two of them produces a non-trivial unitary transformation on the degenerate ground-state manifoldhttps://www.nature.com/articles/npjqi20151.
The primary motivation for studying MZMs and their braiding properties is their combination of non-local encoding and spectral protection. When a fermionic state is formed from two spatially separated MZMs, local environmental noise cannot easily determine or flip the encoded state, because such an error would require coupling to both separated Majorana components. Therefore, local perturbations are strongly suppressed as a source of decoherencehttps://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083. MZMs also give rise to ground-state degeneracies. A system with $2n$ Majorana zero

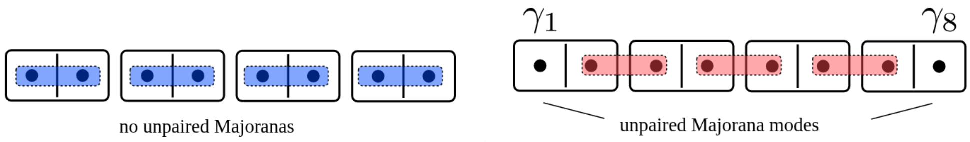

Figure 2.2: Schematic representation of the Kitaev chain in its two distinct phases. Left: In the trivial phase, Majorana modes on the same site pair to form conventional fermions, leaving no unpaired modes. Right: In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends, which combine into a non-local fermionic mode. Topocondmat

modes spans a 2^n-dimensional fermionic Hilbert space associated with the occupations of n non-local fermions. If the total fermion parity is fixed, the physical ground-state degeneracy is reduced to 2^{n-1}. In either case, the useful low-energy manifold must be separated from excited states by a finite spectral gap, which protects the encoded information against thermal excitations.https://arxiv.org/abs/cond-mat/0010440.
The remaining question is how such Majorana operators can emerge from an ordinary fermionic Hamiltonian. The Kitaev chain provides the minimal model where this mechanism can be seen explicitly, and it also gives the conceptual starting point for the Poor Man's Majorana systems studied in this thesis.

2.2 Kitaev Chain and Poor Man's Majorana Systems

Before discussing the Majorana models used in this thesis, we first introduce their theoretical origin: the Kitaev chain. The Kitaev chain, or Kitaev-Majorana chain, is a simplified model for a topological superconductor. First proposed by Alexei Kitaev in 2000 https://arxiv.org/pdf/cond-mat/0010440, the Kitaev chain is a one-dimensional superconducting system that can host Majorana zero modes at its ends under certain conditions, illustrated in Figure 2.2. The model is defined on a lattice of spinless fermions with nearest-neighbor hopping and superconducting pairing. The Hamiltonian of the Kitaev chain is

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_{j=1}^{N-1} \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta \sum_{j=1}^{N-1} \left(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger \right), \tag{2.4}$$

Each term in Eq. 2.4 has a clear physical interpretation. The first term represents the chemical potential μ, which controls the filling of the fermionic states. The second term describes the hopping of fermions between neighboring sites with amplitude t. The third term represents p-wave superconducting pairing, where pairs of fermions are created or annihilated on adjacent sites with amplitude Δ.
As shown schematically in Fig. 2.2, the Kitaev chain has two distinct phases depending on the parameters μ, t, and Δ. In the trivial phase, the Majorana modes on the same site pair up to form conventional fermions, leaving no unpaired modes. In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends. These unpaired Majoranas combine into a non-local fermionic mode that can be occupied or empty without costing energy, leading to a twofold degeneracy in the ground state. This degeneracy is protected as long as local perturbations do not close the bulk gap or directly hybridize the edge modes.

The topological phase of the Kitaev chain is the basic model for the Majorana zero modes studied in this thesis. The goal is then to understand how similar Majorana-like physics can be engineered in more realistic systems.

2.2.1 Bulk Spectrum and Topological Phase

In order to determine when the Kitaev chain is topological, we need to consider the properties of the bulk spectrum.
For a translationally invariant chain, we can perform a Fourier transform to momentum space, which allows us to analyze the energy spectrum and identify the conditions for the topological phase. The Fourier transform is given by

$$c_j = \frac{1}{\sqrt{N}} \sum_p e^{ipaj} c_p \tag{2.5}$$

This transformation allows the Hamiltonian to be expressed in terms of momentum-space operators c_p and $c_p^\dagger$. The resulting Hamiltonian in momentum space is

$$H = \sum_p \xi_p \left(c_p^\dagger c_p - \frac{1}{2} \right) - \sum_p t \cos pa + \Delta \sum_p \left(c_p^\dagger c_{-p}^\dagger e^{ipa} + c_{-p} c_p e^{-ipa} \right), \tag{2.6}$$

with $\xi_p = -\mu - 2t \cos pa$ and a being the lattice spacing. In the Bogoliubov–de Gennes formalism, the quasiparticle spectrum becomes

$$E_{p,\pm} = \pm\sqrt{(\mu + 2t\cos pa)^2 + 4\Delta^2 \sin^2 pa}. \tag{2.7}$$

Even though the Bogoliubov–de Gennes formalism describes a non-interacting quasiparticle problem, it still provides the basic topological picture that the interacting models in this thesis build on. In Fig. 2.3, we can see how the spectrum changes as the chemical potential μ is tuned. The presence of a gap is crucial for the stability of the Majorana modes, and the closing and reopening of this gap signals a topological phase transition.

The quasiparticle spectrum is gapped for most parameter values, but the gap closes when

$$|\mu| = 2|t|.$$

This is also visible in Fig. 2.3. The gap closing marks the transition between the trivial and topological phases. When μ crosses this boundary, the ground state changes its topological character without requiring ordinary symmetry breaking Qtech.

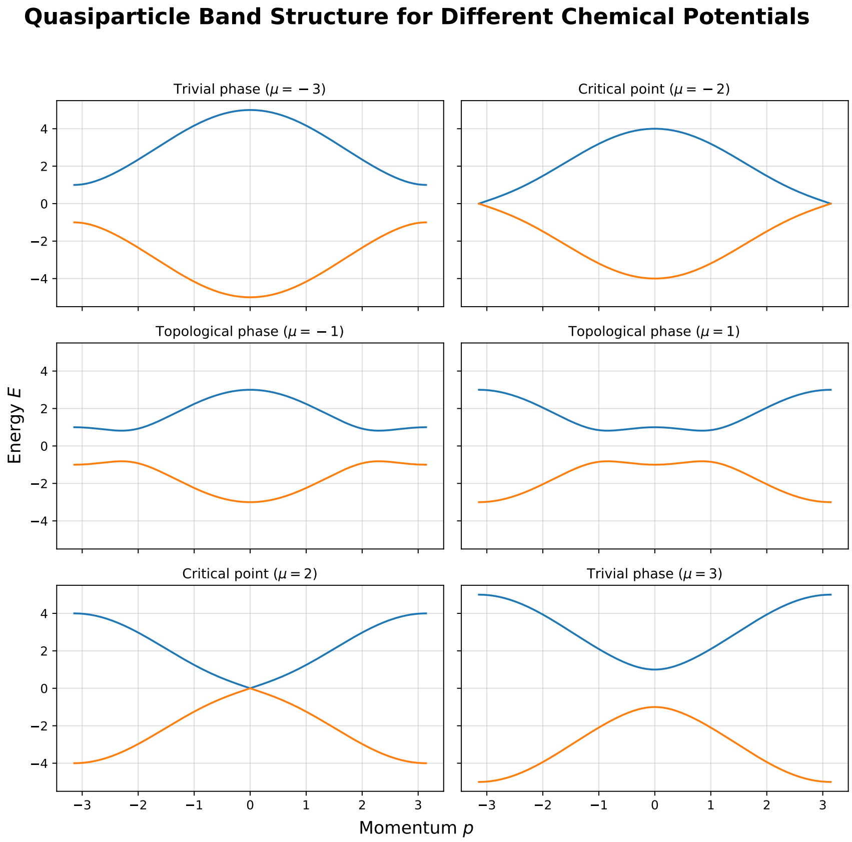

Figure 2.3: Quasiparticle energy spectrum of the Kitaev chain for different chemical potentials μ. Left: In the trivial phase ($|\mu| > 2|t|$), the spectrum is fully gapped with no zero-energy modes. Right: In the topological phase ($|\mu| < 2|t|$), the gap closes and reopens, signaling the emergence of zero-energy edge modes.

2.2.2 Majorana Representation and Zero Modes

Looking back at Fig. 2.2, we want a representation that makes the unpaired edge modes more explicit. To do so, we express the fermionic operators in terms of Majorana operators. For each site j, we define two Majorana operators γ_j^A and γ_j^B as follows:

$$c_j = \frac{1}{2}(\gamma_j^A + i\gamma_j^B), \quad c_j^\dagger = \frac{1}{2}(\gamma_j^A - i\gamma_j^B), \tag{2.8}$$

where $\gamma_j^{A,B}$ satisfy $\{\gamma_j^\alpha, \gamma_k^\beta\} = 2\delta_{jk}\delta_{\alpha\beta}$ and $(\gamma_j^\alpha)^\dagger = \gamma_j^\alpha$. In the trivial phase, the Majorana operators pair on the same site. In the topological phase, and especially at the ideal point $t = \Delta$ and $\mu = 0$, γ_j^B couples to γ_{j+1}^A. This leaves two edge modes unpaired, γ_1^A and γ_N^B, at the chain ends. These two unpaired Majoranas define a non-local fermionic mode,

$$f = \frac{1}{2}(\gamma_1^A + i\gamma_N^B), \tag{2.9}$$

which satisfies $\{f, f^\dagger\} = 1$ and costs zero energy to occupy. This non-local fermionic mode creates the basis for the exploration of Majorana physics in this thesis, and it is the key to understanding how Majorana modes can be used for quantum computation. The occupation of this non-local mode corresponds to a twofold degeneracy in the ground state, which is protected as long as local perturbations do not close the bulk gap or directly hybridize the two edge modes. This non-local encoding of information is what gives Majorana modes their potential for fault-tolerant quantum computation, as local errors cannot easily access or change the encoded state.Qtech

2.2.3 Poor Man's Majorana Systems

The ideal Kitaev chain is a useful theoretical model, but it is not directly realizable in most physical systems, as it ideally prefers to be infinite, homogeneous, and non-interacting. The reason for this, is that the Majorana modes are localized at the edges of the chain, and their energy splitting decreases exponentially with the chain length $\exp\left(-\frac{L}{l_0}\right)$https://arxiv.org/pdf/cond-mat/0010440. This would make the Majorana modes robust against any local perturbation. However, in practice we have to work with finite systems, and the Majorana modes will generally have a small energy splitting due to their overlap. However, a Poor mans Majorana system is a finite system, which under the right parameter conditions, can mimic the low-energy physics of a Kitaev chain, including the presence of Majorana-like modes, in a much more compact and experimentally accessible setup.
While the Kitaev chain is the minimal theoretical model, it assumes spinless fermions and intrinsic p-wave pairing. Real superconductors are usually spinful and dominated by conventional s-wave pairing. The Poor Man's Majorana (PMM) platform is a compact engineered system whose low-energy behavior can mimic a short Kitaev chain. The simplest version is a double quantum dot coupled through a superconducting segment. Each dot acts as a site, elastic cotunneling produces effective hopping, and crossed Andreev reflection produces an effective non-local pairing term.

The effective two-dot Hamiltonian can be written as

$$H = \sum_{i=L,R} \epsilon_i n_i + t(d_L^\dagger d_R + d_R^\dagger d_L) + \Delta(d_L^\dagger d_R^\dagger + d_R d_L) + U n_L n_R, \tag{2.10}$$

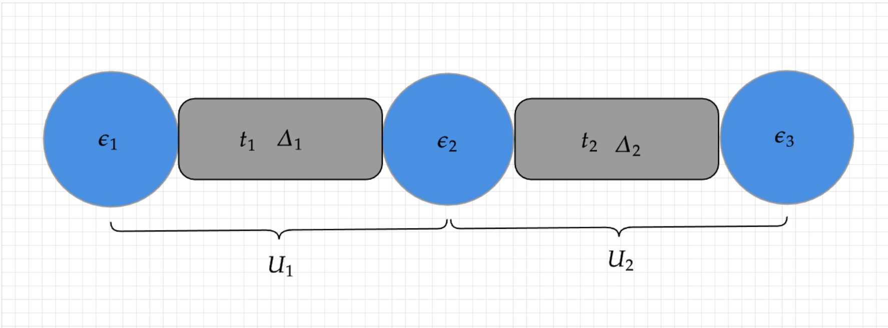

Figure 2.4: Schematic representation of the PMM setup. The three quantum dots are represented by the blue circles, and the superconducting segments are represented by the grey rectangles. The tunable parameters of the system, such as the dot energy levels, ECT, CAR and the Coulomb interaction, are indicated by their respective labels. By tuning these parameters, we can access different regimes of the system and identify the sweet spots where Majorana-like modes emerge.

where $d_{L/R}^{\dagger}$ and $d_{L/R}$ are creation and annihilation operators on the left and right dots, n_i is the number operator, $\epsilon_{L,R}$ are the dot energy levels, t is the elastic cotunneling amplitude, Δ is the crossed Andreev reflection amplitude, and U is the interdot Coulomb interaction energy. The parameters map directly onto the Kitaev-chain language:

- **Elastic cotunneling (ECT):** An electron tunnels coherently from one dot to the other through a virtual process in the superconductor. This produces the effective hopping amplitude t.
- **Crossed Andreev reflection (CAR):** A Cooper pair splits, with one electron entering each dot. This produces the non-local pairing amplitude Δ.
- **On-site energy:** The dot energies ϵ_L and ϵ_R play the role of local chemical potentials.

At the ideal non-interacting sweet spot, $|\epsilon_L| = |\epsilon_R| = 0$ and $|t| = |\Delta|$, the two-dot system develops Majorana-like states localized primarily on opposite dots. Since the system is finite and lacks a true bulk, these states are not topologically protected in the same sense as an infinite Kitaev chain. This is why they are referred to as Poor Man's Majoranas Qtech.

For the purposes of this thesis, the PMM idea is generalized from two dots to interacting n-dot chains. A schematic representation is shown in Figure 2.4. Adding more dots introduces more tunable parameters and more possible low-energy structure, but it also makes the analytic sweet-spot problem much harder. This motivates the numerical optimization strategy discussed later in the thesis.

2.3 Diagnostics of Majorana-like States

In a finite PMM system, the presence of low-energy or near-zero-energy states is necessary, but not sufficient, evidence for Majorana physics. As discussed in the previous section, the ideal Kitaev chain has zero-energy Majorana modes at its ends, producing a degenerate low-energy manifold. In a short and finite PMM chain, however, the Majorana modes can overlap, hybridize, and split away from zero energy. Other ordinary low-energy states can also appear because the dot energies, tunnel couplings, pairing amplitudes, and interactions are all tunable.

For this reason, the identification of Majorana-like states must rely on several complementary diagnostics rather than a single spectral feature. The diagnostics used here can be divided into two broad groups. Spectral diagnostics test whether the low-energy manifold has the expected parity degeneracy and excitation gap. Local diagnostics test whether the corresponding states are non-local, locally indistinguishable, and Majorana-like in their particle-hole structure. Together, these quantities define what is meant in this thesis by a promising Majorana-like configuration.

2.3.1 Fermion Parity

The first diagnostic is fermion parity. The mean-field Hamiltonian of a superconductor does not conserve particle number, because electrons can be added or removed through Cooper-pair processes. It does, however, conserve the parity of the total particle number. The parity operator can be written as

$$P = (-1)^N = \prod_n (1 - 2c_n^\dagger c_n),$$

where N is the total particle number. The eigenvalues of P are $+1$ for even parity and -1 for odd parity.

In a PMM system, the qubit is encoded in the global fermion parity of the non-local fermionic mode formed by the two Majorana components. At the ideal two-dot sweet spot, where the effective hopping and pairing satisfy $t = \Delta$, the ground state becomes twofold degenerate, with one state in the even-parity sector and one in the odd-parity sector https://www.nature.com/articles/s41586-022-05585-1; https://doi.org/10.1103/PhysRevB.86.134528. Classifying the spectrum by parity is therefore the first step in identifying candidate Majorana-like states. In this thesis, this parity classification is also used to ask a more detailed question: whether even-odd degeneracy can persist beyond the ground state in finite interacting chains, and what such higher-level degeneracy would imply for robustness.

2.3.2 Energy Degeneracy and Spectral Gap

Once the states have been separated into parity sectors, the next diagnostic is the energy splitting between parity partners. Majorana-like physics requires a near-degenerate pair of states, usually one even and one odd, associated with the occupation of the non-local fermionic mode. In an ideal infinite Kitaev chain, this degeneracy is exact. In a finite PMM chain, the edge modes can overlap, producing a small but nonzero splitting. The goal is therefore not only to find low-energy states, but to find parameter regimes where the even-odd splitting is small compared with all other relevant energy scales.

The excitation gap is equally important. The gap separates the low-energy parity manifold from quasiparticle excitations and suppresses thermal or dynamical leakage out of the computational subspace. In PMM and quantum-dot-chain proposals, the

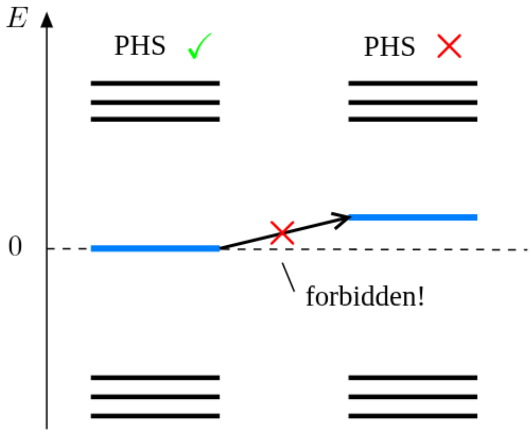

Figure 2.5: Energy spectrum showing how the energy eigenvalues of the system must be symmetric around zero energy due to particle-hole symmetry. Topocondmat

useful gap is controlled by the same physical couplings that create the effective Kitaev chain, especially the elastic cotunneling and crossed-Andreev pairing amplitudes. If the gap closes, the Majorana-like signature is no longer protected, and the low-energy interpretation becomes unreliable https://www.nature.com/articles/ncomms1966. A sizable gap is also essential for the braiding calculations in this thesis, because it justifies projecting the dynamics onto a low-energy subspace.

This thesis also explores whether finite interacting chains can display approximate even-odd degeneracy across several energy levels, not only in the ground state. If such a structure survives, then some Majorana-like features may remain meaningful even when excited states are included in the projected Hilbert space. Figure 2.5 illustrates the particle-hole symmetric structure expected in the idealized superconducting spectrum. In practice, the relevant diagnostic is whether the candidate parity partners are both nearly degenerate and well separated from the rest of the spectrum Topocondmat; https://www.nature.com/articles/s41586-022-05585-1.

2.3.3 Local Distinguishability

A defining feature of spatially separated Majorana modes is that the encoded information is non-local. Local measurements should therefore not reveal whether the system is in the even or odd ground state. This principle is central to the topological motivation for Majorana qubits: the qubit state is stored globally, while local perturbations should have only limited access to it https://doi.org/10.1103/RevModPhys.80.1083. In the PMM setting, the same idea appears in a finite-system form: the parity qubit can only be fully read out by probing both Majorana components, rather than a single dot alone https://doi.org/10.1103/PhysRevB.86.134528.

One way to quantify this property is through the local distinguishability parameter

$$\mathrm{LD} = \frac{1}{N}\sum_{j=1}^{N} ||\rho_j^o - \rho_j^e|| = \frac{1}{N}\sum_{j=1}^{N} ||\delta\rho_j||.$$

Lund1; VikMar; https://doi.org/10.1103/PhysRevB.110.155436. Here, $\rho_j^{o/e}$ are the reduced density matrices of site j for the odd and even parity ground states, respectively, and $||\cdot||$ is the trace norm. A value close to zero indicates that the two states are locally indistinguishable, while a larger value indicates that local measurements can tell the two parity states apart.

2.3.4 Majorana Polarization

Local indistinguishability tests whether the information is hidden from local measurements, but it does not by itself say whether the operator connecting the two parity states has the particle-hole structure expected of a Majorana mode. For this, we use the Majorana polarization (MP). The MP is a local measure of how Majorana-like a state is: an ideal Majorana mode should be self-conjugate, with balanced particle and hole character, and should be localized near the ends of the chain. In an ideal PMM configuration, the polarization should therefore be concentrated at the edge dots, with opposite signs on the two ends and little or no weight in the bulk https://doi.org/10.1103/PhysRevB.110.155436.

One useful definition is

$$M_\alpha = \frac{W_\alpha^2 - Z_\alpha^2}{W_\alpha^2 + Z_\alpha^2},$$

where

$$W_\alpha = \sum_\sigma \langle o| c_{\alpha\sigma} + c_{\alpha\sigma}^\dagger |e\rangle\,, \quad Z_\alpha = i\sum_\sigma \langle o| c_{\alpha\sigma} - c_{\alpha\sigma}^\dagger |e\rangle\,.$$

Here, α denotes the site index, σ is the spin index, and $|o\rangle$ and $|e\rangle$ are the odd- and even-parity ground states. In an ideal sweet spot with two perfectly localized Majoranas, the edge values would satisfy $M_L = -M_R = \pm 1$ Qtech. In the spinless model used later in this thesis, the same idea is applied without the spin sum: the diagnostic checks whether the parity-changing Majorana matrix elements are localized at the chain ends with the expected relative sign.

2.3.5 Charge Expectation and Non-locality

Another diagnostic is the local charge expectation value in the even and odd states,

$$\langle e| n_{L(R)} |e\rangle\,, \quad \langle o| n_{L(R)} |o\rangle\,.$$

An ideal Majorana mode has no ordinary local charge identity: it is an equal particle-hole superposition rather than a conventional electron or hole. In a PMM system, this means that the local charge distribution should not reveal the parity state of the non-local fermion. At the sweet spot, the relevant charge signatures become symmetric between the two parity states, corresponding to the “half-electron” or charge-neutral character of the Majorana components [4, 3].

For the diagnostics used here, the important quantity is not the absolute charge on a dot, but the difference between the even- and odd-parity partners. A small charge difference means that the two states share nearly the same local charge distribution,

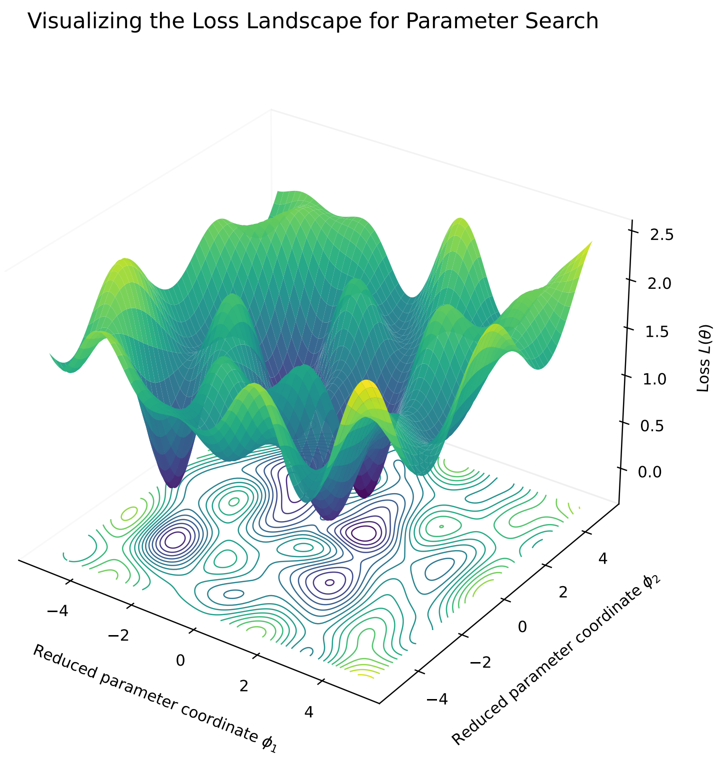

Figure 2.6: Visualization of the parameter space for an n-dot system, showing how multiple parameters and their combinations create a complex landscape with multiple local minima.

which is another way of saying that the parity information is stored non-locally rather than being visible on a single dot.

2.3.6 Optimization Motivation

Together, these diagnostics provide the physical ingredients used later to build the optimization loss function. For a simple two-dot system, the relevant tunable parameters are the dot energies, elastic cotunneling, crossed Andreev reflection, and interaction strength. For longer chains, the number of tunable parameters grows quickly, visualized in Figure 2.6. Since interactions and inhomogeneous parameters make it difficult to identify sweet spots analytically, the diagnostics above can be combined into a loss function and used to search numerically for promising Majorana-like regimes. The resulting loss landscape may contain many local minima, which motivates the repeated-restart and basin-hopping strategy described in the methods chapter.

2.4 Majorana Exchange and Braiding

Braiding is the operation that makes Majorana modes useful for topological quantum computation. The basic idea is to exchange two Majorana modes adiabatically so that the quantum state transforms according to their non-Abelian statistics. Because the transformation is tied to the topology of the exchange, rather than to the exact microscopic path, braiding offers a route to gates that are less sensitive to local control errors. Braiding in the right patterns, with the right systems, can lead to the implementation of useful quantum gates.SOURCE.
In a system of Poor Man's Majoranas, braiding does not come from physically moving the Majorana modes or any other physical pieces of the setup, but rather by tuning the parameters of the system in a way that mimics the effect of an exchange. What this does, is couple the Majorana modes in a way that produces the same unitary transformation on the low-energy subspace as a physical braid would. This is sometimes called "measurement-only" or "parameter-tuning" braiding, and it relies on the same underlying topological principles as physical braiding, but it is implemented through a sequence of parameter changes rather than through spatial motion.

2.4.1 From Qubit Gates to Fermionic Operators

In quantum computation, quantum gates are unitary operations that manipulate the state of qubits. Quantum gates, or quantum logic gates, are the building blocks of quantum circuits, the same way classical logic gates are for digital circuits. Unlike many classical logic gates, quantum logic gates are reversible, meaning that they can be undone by applying the inverse operation. This reversibility is a fundamental property of quantum mechanics and is essential for quantum computation.

Quantum gates are unitary operations and can be described as unitary matrices, relative to some orthonormal basis. For example, the Pauli gates X, Y, and Z are represented by the following matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.11}$$

The Pauli gates (X, Y, Z) are fundamental single-qubit gates that correspond to rotations around the x, y, and z axes of the Bloch sphere, respectively. They are also examples of involutory gates, meaning that applying them twice yields the identity operation (e.g., $X^2 = Y^2 = Z^2 = I$).

In order to replicate these quantum logic gates using Majorana modes, we need to understand how to express these gates in terms of fermionic operators. This is because the Majorana modes are described by fermionic operators, and we want to find a way to translate the abstract quantum gates into operations that can be implemented in a Majorana system. By expressing the Pauli gates in terms of fermionic creation and annihilation operators, we can identify how to manipulate the Majorana modes to achieve the desired quantum gate operations. This connection is crucial for implementing quantum computation using Majorana-based qubits.
Let us start by introducing two quantum states $|0\rangle$ and $|1\rangle$, which we can simply represent

as column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.12}$$

These two vectors represent the two possible states of a qubit, where $|0\rangle$ is the state where the qubit is in the "0" state and $|1\rangle$ is the state where the qubit is in the "1" state. These states form the computational basis for a single qubit. The Pauli gates act on these basis states as follows:

$$\begin{aligned}
X\,|0\rangle &= |1\rangle\,, \quad X\,|1\rangle = |0\rangle\,, \\
Y\,|0\rangle &= i\,|1\rangle\,, \quad Y\,|1\rangle = -i\,|0\rangle\,, \\
Z\,|0\rangle &= |0\rangle\,, \quad Z\,|1\rangle = -\,|1\rangle\,.
\end{aligned}$$

To put these operations in the context of fermionic systems, we can see that the X gate corresponds to a bit-flip operation, or in other words, it annihilates the "0" state and creates the "1" state, and vice versa. The Y gate also flips the states but introduces a phase factor of i or $-i$, meaning it combines a bit-flip with a phase shift. The Z gate, on the other hand, does not change the state but applies a phase of -1 to the $|1\rangle$ state while leaving the $|0\rangle$ state unchanged.

Putting these operations in the context of fermionic operators, we can express the Pauli gates in terms of creation and annihilation operators for a two-site system. For example, if we have fermionic operators c_1 and c_2 corresponding to two sites, we can write:

$$\begin{aligned}
X &= c_1^\dagger c_2 + c_2^\dagger c_1, \\
Y &= -i(c_1^\dagger c_2 - c_2^\dagger c_1), \\
Z &= c_1^\dagger c_1 - c_2^\dagger c_2.
\end{aligned} \tag{2.13}$$

We can now see how easily the quantum logic gates can be transformed into fermionic language, which is the first step towards understanding how to implement these gates in a Majorana system. By expressing the gates in terms of fermionic operators, we can identify the necessary interactions and couplings in a Majorana-based setup that would allow us to perform the desired quantum gate operations. This is crucial for the development of Majorana-based quantum computation, as it provides a clear pathway for translating abstract quantum operations into physical manipulations of Majorana modes.

2.4.2 Jordan-Wigner Transformation

Jordan-Wigner transformation is a mathematical technique used to map spin operators to fermionic creation and annihilation operators. Proposed by Pascual Jordan and Eugene Wigner in 1928https://link.springer.com/article/10.1007/BF01331938, where they originally introduced it to solve the one-dimensional spin-1/2 chain problem, but later was extended to two-dimensional systems and beyond. The transformation is particularly useful in the study of quantum many-body systems, as it allows for the analysis of spin systems using fermionic techniques.
The Jordan-Wigner transformation is useful here because it lets us represent qubit or spin operators in terms of fermionic operators while preserving the correct anticommutation

relations between different sites. This is not only a formal point: in the numerical braiding calculations, Majorana operators from different sites and subsystems must be embedded into a common Hilbert space with the correct fermionic algebra. The Jordan-Wigner strings provide this bookkeeping. The transformation allows us to express the spin-1/2 Pauli operators in terms of fermionic creation and annihilation operators$https://futureofmatter.com/assets/fermions_and_jordan_wigner.pdf$. For a chain of spin-1/2 particles, the transformation is given by:

$$
\begin{aligned}
\sigma_j^+ &= \frac{1}{2}(\sigma_j^x + i\sigma_j^y) \equiv f_j^\dagger, \\
\sigma_j^- &= \frac{1}{2}(\sigma_j^x - i\sigma_j^y) \equiv f_j, \\
\sigma_j^z &= 2f_j^\dagger f_j - 1,
\end{aligned}
$$

Leaving us with the correct same-site fermionic relations, but operators on different sites commute instead of anticommute. To fix this, we take a chain of fermions, and define a new set of fermionic operators:

$$
\begin{aligned}
a_j^\dagger &= e^{i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j^\dagger \\
a_j &= e^{-i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} \cdot f_j, \\
a_j^\dagger a_j &= f_j^\dagger f_j
\end{aligned}
$$

The exponential term, can be simplified to a string of σ^z operators, giving us

$$
e^{\pm i\pi \sum_{k=1}^{j-1} f_k^\dagger f_k} = \Pi_{k=1}^{j-1}(-\sigma_k^z). \tag{2.14}
$$

The transformed spin operators now satisfy the correct fermionic anticommutation relations:

$$
\begin{aligned}
\{a_j^\dagger, a_k\} &= \delta_{jk}, \\
\{a_j^\dagger, a_k^\dagger\} &= 0, \\
\{a_j, a_k\} &= 0.
\end{aligned}
$$

The Jordan-Wigner transformation is a powerful tool for analyzing spin systems using fermionic techniques, and it provides a clear example of how quantum gates can be expressed in terms of fermionic operators. By applying this transformation, we can gain insights into the behavior of spin systems and their connections to fermionic systems, which is essential for understanding the physics of Majorana modes and their applications in quantum computation.

2.4.3 Pauli Matrices in the Majorana Representation

The reason for rewriting the Pauli operators in terms of Majorana bilinears is that braiding acts naturally on Majorana operators, while the gate implemented by the braid is usually compared with a Pauli operation on an encoded qubit. By expressing the Pauli gates in terms of Majorana bilinears, we can directly relate the physical operations performed on the Majorana modes to the abstract quantum gates that we want to implement. This connection is important to both understand and implement quantum gates in a Majorana-based quantum computing platform. We start by considering four Majorana operators, grouped into two ordinary fermions:

$$\begin{aligned} c_1 &= \frac{1}{2}(\gamma_0 + i\gamma_1), \quad c_1^\dagger = \frac{1}{2}(\gamma_0 - i\gamma_1), \\ c_2 &= \frac{1}{2}(\gamma_2 + i\gamma_3), \quad c_2^\dagger = \frac{1}{2}(\gamma_2 - i\gamma_3). \end{aligned} \tag{2.15}$$

These four Majorana operators γ_0, γ_1, γ_2, and γ_3 span a four-dimensional Hilbert space, which can be thought of as the state space of two fermionic modes. By fixing the total fermion parity, we can select a two-dimensional subspace that serves as the computational basis for a qubit. Within such a fixed-parity subspace, we can represent the Pauli operators as bilinears of the Majorana operators. Specifically, we can write:

$$\begin{aligned} X &= i\gamma_0\gamma_3 = -i\gamma_1\gamma_2, \\ Y &= -i\gamma_0\gamma_2 = -i\gamma_1\gamma_3, \\ Z &= i\gamma_0\gamma_1 = i\gamma_2\gamma_3. \end{aligned}$$

These expressions can be derived by substituting the fermionic operators in Eq. 2.13 with the Majorana representation of the fermionic operators. The resulting bilinears of Majorana operators correspond to the Pauli gates acting on the encoded qubit.
Thus, controlled Majorana couplings of the form $i\gamma_i\gamma_j$ act as Pauli operations on the encoded qubit. This is the operator-level connection between Majorana braiding and gate implementation. https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083; https://arxiv.org/abs/1202.1293; https://scipost.org/SciPostPhys.3.3.021

2.4.4 Majorana Exchange Gates

An anticlockwise exchange of two Majoranas γ_i and γ_j is represented by the unitary

$$B_{ij} = \exp\left(-\frac{\pi}{4}\gamma_i\gamma_j\right) = \frac{1}{\sqrt{2}}(1 - \gamma_i\gamma_j),$$

up to orientation conventions and an overall phase. With this convention, the unitary acts on the Majorana operators as

$$\begin{aligned} B_{ij}\gamma_i B_{ij}^\dagger &= \gamma_j, \\ B_{ij}\gamma_j B_{ij}^\dagger &= -\gamma_i, \\ B_{ij}\gamma_k B_{ij}^\dagger &= \gamma_k, \quad k \neq i, j. \end{aligned}$$

The important point is that exchanges are generated by Majorana bilinears. Therefore, the same $i\gamma_i\gamma_j$ operators that represent Pauli operations in the encoded qubit space also determine the target braid gates used later in this thesis. For example, squaring an exchange produces a parity-dependent bilinear operation,

$$B_{ij}^2 = -\gamma_i\gamma_j,$$

up to phase and orientation conventions. This is why the numerical braid unitary can be checked by comparing its action on the Majorana operators and by comparing its projected parity blocks with the expected target gate. https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.80.1083; https://arxiv.org/abs/1202.1293

The remaining question is how to compute the unitary generated by an adiabatic coupling protocol, which is the role of Kato transport in the next section.

2.5 Projected Subspaces, Kato Transport and Braid Unitaries

This section connects the high-level topological theory of braiding with the microscopic details of the PMM system. The goal is to understand how to compute the braid unitary generated by a time-dependent parameter protocol, and how to compare it with the ideal target gate. In these systems, braiding is not performed by physically moving particles through space, but rather by tuning Hamiltonian parameters in a way that mimics the effect of an exchange.
An adiabatic parameter protocol defines a path in the space of Hamiltonians. If the relevant low-energy subspace remains isolated along this path, the geometric transport of that subspace gives the braid unitary. The Kato formulation provides a convenient way to compute this unitary without having to track individual eigenstates and their arbitrary phases. By projecting the dynamics onto the low-energy subspace defined by the Majorana-like states, we can extract the effective braid unitary and compare it with the ideal target gate expressed in terms of Majorana bilinears.

2.5.1 Kato Transport in a Degenerate Subspace

The Berry phase is a geometric phase acquired by a quantum state during adiabatic evolution around a closed loop in parameter space. For a non-degenerate eigenstate, the state returns to itself up to a phase. This phase has a dynamical part, which depends on the time scale and the energy of the state, and a geometric part, which depends only on the path in parameter space. The geometric contribution is the Berry phase.

For a degenerate subspace, the situation is richer. The adiabatic evolution can mix states within the degenerate manifold, so the Berry phase becomes a matrix rather than a scalar phase. In this thesis, this non-Abelian Berry matrix is what determines the braid unitary acting on the Majorana qubit space.

The Kato formulation provides a convenient way to compute this adiabatic transport. Instead of tracking individual eigenvectors and their phases, one works directly with the projector $P(t)$ onto the relevant low-energy subspace. In the convention used in the numerical implementation, the Kato evolution satisfies

$$\partial_t U_t = [P, \partial_t P]\, U_t, \qquad U_0 = I.$$

https://journals.jps.jp/doi/epdf/10.1143/JPSJ.5.435 The final unitary U_T is the geometric transport associated with the path in the chosen subspace. A full physical time evolution would also contain dynamical phases depending on the instantaneous energies and total protocol time. The Kato construction isolates the adiabatic geometric part, which is the component relevant for identifying the braid gate. In numerical calculations, $P(t)$ is obtained by diagonalizing the Hamiltonian at each time step and constructing the projector onto the chosen low-energy manifold.

This unitary matrix is the braid unitary generated by the parameter protocol. By comparing U_T with the ideal target gate expressed in terms of Majorana bilinears, we can assess how well the parameter-tuning braid approximates the desired quantum gate operation. This comparison can be done both at the operator level, by checking how U_T

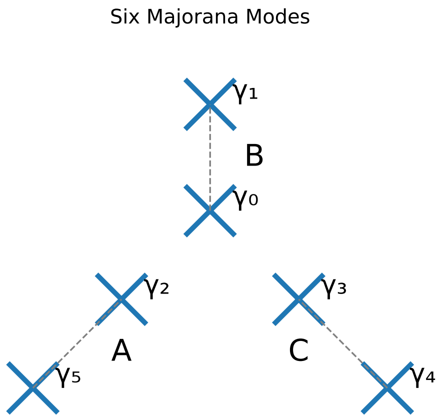

Figure 2.7: Schematic representation of a system hosting six majoranas. The illustrations shows pairs of two and two Majoranas coupling together, in a Y-junction, each pair form a subsystem that can be described by a Poor Man's Majorana setup, A, B and C.

transforms the Majorana operators, and at the level of the projected parity blocks, by checking how U_T acts on the encoded qubit states.

2.5.2 Effective Braiding Hamiltonian

As a reference model, assume that the system contains four Majorana modes γ_0, γ_1, γ_2, and γ_3 involved in the braid. A useful physical picture is a Y-junction made from three PMM subsystems, where one central Majorana is coupled sequentially to different neighboring Majoranas.

In Fig. 2.7, the four Majoranas γ_0, γ_1, γ_2, and γ_3 are the modes actively used in the braid, while γ_4 and γ_5 form a spectator pair. The ideal effective Hamiltonian can be written as

$$
\begin{aligned}
H = &\Delta_{01} i\gamma_0\gamma_1 + \\
&\Delta_{02} i\gamma_0\gamma_2 + \\
&\Delta_{03} i\gamma_0\gamma_3,
\end{aligned}
\tag{2.16}
$$

where Δ_{ij} is a tunable coupling between Majoranas γ_i and γ_j. By turning these couplings on and off in a controlled sequence, the system can implement an effective exchange operation. This model is not meant to describe all microscopic details of the dot device directly. Instead, it serves as a reference model where the intended Majorana exchange can be tested without additional complications from the full interacting Hamiltonian.

2.5.3 Projected Microscopic Junctions

In a microscopic PMM device, the tunable quantities are not abstract Majorana bilinears, but physical hopping and pairing couplings between dots or subsystems. A junction between two PMM subsystems can be modeled by

$$
H_{JuncAB} = t_{AB}(d_A^\dagger d_B + d_B^\dagger d_A) + \Delta_{AB}(d_A^\dagger d_B^\dagger + d_B d_A), \tag{2.17}
$$

which is the same type of elastic cotunneling and crossed-Andreev pairing structure appearing in the PMM Hamiltonian. After such microscopic junction operators are projected into the retained low-energy subspace, they can be compared with the effective Majorana bilinears of Eq. 2.16. This motivates comparing two related braid constructions: one built from ideal or energy-level Majorana operators, and one built from projected microscopic local operators.

2.5.4 Energy-Level Versus Local-Operator Braids

The central comparison in the braiding analysis is between two ways of replacing the ideal operators γ_2 and γ_3 in Eq. 2.16. The effective Hamiltonian assumes that the Majorana operators are ideal and perfectly localized, while the optimized PMM system only provides approximate Majorana-like degrees of freedom. To test how close the microscopic system comes to the ideal exchange picture, we compare an energy-level construction with a local-operator construction.

In Majorana language, the target exchange operation used in this thesis is

$$B_{\gamma_2\gamma_3} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \tag{2.18}$$

The resulting braid unitary is later compared with this target gate, both through its action on Majorana operators and through its projected parity-block representation.

The local-operator braid is built from microscopic endpoint operators rather than from idealized Majorana operators. In principle, one can project either the real or imaginary local Majorana component, corresponding to $c^\dagger + c$ and $i(c^\dagger - c)$, respectively. In the calculations presented here, the physical braid uses the projected minus component, $P\,i(c^\dagger - c)\,P$, for the relevant B and C subsystem sites. After projection into an N-dimensional low-energy subspace, these operators are normalized and used as the effective braid channels that replace γ_2 and γ_3. The driven Hamiltonian therefore takes the schematic form

$$\begin{aligned} H_{\text{local}} = &\Delta_{01} i\gamma_0\gamma_1 + \\ &\Delta_{02} i\gamma_0\Gamma_B^P + \\ &\Delta_{03} i\gamma_0\Gamma_C^P, \end{aligned} \tag{2.19}$$

with

$$\Gamma_B^P \propto P\,i(c_B^\dagger - c_B)\,P, \qquad \Gamma_C^P \propto P\,i(c_C^\dagger - c_C)\,P. \tag{2.20}$$

Mention that it is physically meaningful to check both the X and Y local Majorana components.

A local fermionic operator has two Majorana quadratures,

$$\chi_x = c^\dagger + c, \qquad \chi_y = i(c^\dagger - c).$$

Choosing only χ_y assumes that the physically relevant Majorana is aligned with that quadrature. But in an interacting/projected low-energy system, the effective Majorana can be rotated in this local Majorana plane,

$$\chi(\theta) = \cos\theta\,\chi_x + \sin\theta\,\chi_y.$$

Here P is constructed by diagonalizing the full Hamiltonian and selecting the N lowest-energy eigenstates. The local-operator braid therefore tests whether the physically projected microscopic endpoint operators access the same exchange channel as the ideal target gate.

The energy-level construction starts from the optimized subsystem Hamiltonian. For clarity, this is the n-dot extension of the effective two-dot model 2.10,

$$H = \sum_{i}^{n} \epsilon_i n_i + \sum_{j}^{n-1} t_j (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) + \sum_{j}^{n-1} \Delta_j (c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j) + \sum_{j}^{n-1} U n_j n_{j+1}, \quad (2.21)$$

Diagonalizing this Hamiltonian 2.21, we obtain the energy eigenstates, which can be classified by parity. By pairing even and odd states that are close in energy, we can construct effective Majorana operators as combinations of these parity partners. In its simplest form, this construction looks like:

$$\begin{aligned} \gamma_+ &= \sum_{i}^{n} \left(|e_i\rangle \langle o_i| + |o_i\rangle \langle e_i| \right), \\ \gamma_- &= i \sum_{i}^{n} \left(|e_i\rangle \langle o_i| - |o_i\rangle \langle e_i| \right). \end{aligned} \quad (2.22)$$

Here n is the number of even-odd energy pairs included in the construction https://arxiv.org/pdf/2510.20538. For $n = 1$, only the ground-state parity pair contributes. Larger values of n retain more of the low-energy spectrum and allow the braid construction to be tested in enlarged projected spaces. The paired-state construction is designed to realize the Majorana algebra within the selected parity-pair subspace; after embedding and projection into the full retained low-energy space, the resulting operators are checked numerically for normalization and anticommutation errors.

In the higher-energy analysis, the relevant projected Majorana channel is not assumed to be purely γ_+ or purely γ_-. Instead, the implementation constructs the accumulated projected γ_+ and γ_- components from the chosen parity pairs, and then fits a global linear combination to the projected local reference operator:

$$\gamma_{\text{fit}} = a\, \gamma_+^P + b\, \gamma_-^P. \quad (2.23)$$

The coefficients a and b are chosen to minimize the mismatch with the projected microscopic local operator in the retained subspace. This fitting procedure can be interpreted as selecting the effective Majorana direction that best aligns the energy-level construction with the physically projected local channel.

The comparison between these two constructions is one of the central checks in this thesis. If the energy-level and local-operator braids agree, then the optimized spectrum and the physical endpoint couplings select the same Majorana exchange channel. If they differ, the spectrum may still contain Majorana-like parity pairs, but the local microscopic operators do not access the ideal braid cleanly. This distinction is what allows the braiding analysis to go beyond asking whether Majorana-like states exist, and instead test whether they can be used to implement the intended exchange operation. odd

Chapter 3

Method

This chapter describes the methods used to optimize the Majorana modes in the junction setup and to analyze their properties. We first discuss the parameter optimization process, including the model parameterization, loss function design based on Majorana criteria, and the optimization strategy. Next, we detail the post-optimization analysis of the setup, covering Hamiltonian reconstruction, parity classification, and various diagnostics. Finally, we describe the braiding protocol implemented in an effective Majorana model and how we validate the results using gate-validation checks.

3.1 Numerical PMM Hamiltonian and Basis

The numerical calculations are built from the spinless n-dot PMM Hamiltonian introduced in the theory chapter. This section specifies how that model is represented in the code: which Hamiltonian is diagonalized, which basis is used, how the fermionic operators are constructed, and how the model parameters are organized before optimization and braiding calculations. The goal here is not to rederive the PMM model, but to define the concrete numerical object used throughout the rest of the method chapter.

3.1.1 Single-chain Hamiltonian

For a single n-dot PMM chain, the Hamiltonian used in the optimization and diagnostic calculations is

$$H_{\mathrm{PMM}} = \sum_{i=1}^{n} \epsilon_i n_i + \sum_{i=1}^{n-1} \left[-t_i \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) + \Delta_i \left(c_i c_{i+1} + c_{i+1}^\dagger c_i^\dagger \right) + U_i n_i n_{i+1} \right]. \quad (3.1)$$

Here $c_i^\dagger$ and c_i create and annihilate a spinless fermion on dot i, and $n_i = c_i^\dagger c_i$ is the corresponding number operator. The onsite energies ϵ_i play the role of local chemical potentials, t_i are elastic cotunneling amplitudes, Δ_i are crossed-Andreev pairing amplitudes, and U_i are nearest-neighbor density-density interaction strengths. This is the same operator structure used both when searching for optimized PMM chains and when reconstructing selected chains for later braiding calculations.

3.1.2 Many-body Fock basis

All operators are represented in the full many-body Fock basis. For a chain with n dots, a basis state is specified by the occupation pattern

$$|n_1 n_2 \cdots n_n\rangle, \qquad n_i \in \{0, 1\}, \tag{3.2}$$

so the Hilbert-space dimension of a single chain is 2^n. Diagonalizing Eq. 3.1 in this basis gives the full many-body spectrum and eigenstates. This is important because the diagnostics used later, such as parity classification, charge expectation values, and Majorana-polarization matrix elements, are all many-body quantities.

For braiding calculations, several optimized chains are combined into a larger Hilbert space. If D identical PMM subsystems are used, each containing n dots, the total number of fermionic sites is $N_{\text{tot}} = Dn$ and the full Hilbert-space dimension is $2^{N_{\text{tot}}}$. In the three-arm braiding geometry used later, $D = 3$, corresponding to subsystems A, B, and C.

3.1.3 Jordan-Wigner construction of fermionic operators

The fermionic creation and annihilation operators must satisfy the canonical anticommutation relations

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \tag{3.3}$$

In the code, these operators are constructed with Jordan-Wigner strings. Starting from local spin raising and lowering operators on site j, the fermionic operators are represented schematically as

$$c_j^\dagger = \left[\prod_{k<j}(-\sigma_k^z)\right]\sigma_j^+, \qquad c_j = \left[\prod_{k<j}(-\sigma_k^z)\right]\sigma_j^-. \tag{3.4}$$

The string of σ^z operators keeps track of the fermionic sign picked up when operators on different sites are exchanged. This same construction is used for single-chain calculations and for the enlarged three-subsystem Hilbert space. In the latter case, the site ordering fixes the Jordan-Wigner strings across subsystems, ensuring that operators belonging to different PMM chains still anticommute correctly.

Once the creation and annihilation operators have been constructed, the code precomputes the basic operator products needed repeatedly:

$$n_i = c_i^\dagger c_i, \qquad h_{ij} = c_i^\dagger c_j + c_j^\dagger c_i, \qquad p_{ij} = c_i^\dagger c_j^\dagger + c_j c_i, \qquad d_{ij} = n_i n_j. \tag{3.5}$$

These building blocks are then combined with the parameter values in Eq. 3.1 to assemble the Hamiltonian matrix. The same Jordan-Wigner ordering is also used when several PMM chains are embedded into the combined braiding Hilbert space, so that the fermionic operators of different subsystems anticommute correctly. For the three-subsystem geometry, sites are ordered as

$$A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_n, C_1, C_2, \ldots, C_n.$$

Operators acting on subsystem B therefore carry a Jordan-Wigner string over all sites in subsystem A, while all operators acting on subsystem C carry a string over all sites in both A and B. This ensures that the fermionic algebra is preserved across the entire device, which is crucial for correctly defining the Majorana operators and their braiding properties later on.

3.1.4 Parameter conventions

The numerical implementation allows each parameter family in Eq. 3.1 to be fixed, homogeneous, or inhomogeneous. A fixed parameter is held at a prescribed value and is not included in the optimization vector. A homogeneous parameter has the same value on every site or bond, while an inhomogeneous parameter is allowed to vary independently from site to site or bond to bond. This convention makes it possible to compare restricted ansätze with more flexible parameter searches using the same Hamiltonian-building routine.

In the main optimization runs used in this thesis, the hopping amplitudes are fixed to $t_i = 1$, and the interaction strength is fixed to selected values of U. The onsite energies ϵ_i and pairing amplitudes Δ_i are optimized. Fixing the hopping scale prevents the optimizer from lowering the loss through a trivial global rescaling of the kinetic energy, and instead forces the search to find nontrivial onsite and pairing profiles for each interaction regime. This also rules out the possibility of blowing up all other parameters in order to suppress the relative effect of interactions, which would be an unphysical way to achieve a low loss in the strongly interacting regime. The optimized parameters are stored in a reduced vector form, where only the free parameters are included. After optimization, the full set of physical parameters is reconstructed by reinserting fixed values and expanding homogeneous parameters as needed.

After optimization, the selected physical parameters are stored and later reloaded for post-optimization diagnostics and braiding calculations. The same parameter convention is also used when constructing the three-subsystem braiding geometry: the optimized single-chain parameters define each subsystem, while additional inter-subsystem couplings are introduced only in the braiding-model sections below.

3.2 Parameter Optimization and Loss Function

The purpose of the optimization is to find parameter regimes where the finite PMM chain has the spectral and local signatures of Majorana-like states. The theory chapter defined the physical criteria; here they are turned into a numerical objective. For each trial parameter vector, the Hamiltonian in Eq. 3.1 is constructed, diagonalized, separated into parity sectors, and assigned a scalar loss. The optimizer then updates only the free parameters, while fixed parameters are reinserted when the physical Hamiltonian is rebuilt.

3.2.1 Optimization vector and physical parameters

The optimizer does not act directly on a full dictionary of physical parameters. Instead, it acts on a reduced real vector θ containing only the entries that are not fixed by the chosen parameter configuration. For example, if t_i is fixed and U_i is fixed to a selected interaction strength, then those entries are omitted from θ. If a parameter is homogeneous, only one number is included; if it is inhomogeneous, one number is included for each site or bond.

The reduced vector is converted back into a parameter dictionary before the Hamiltonian is evaluated. Schematically,

$$\theta \longrightarrow \{t_i, U_i, \epsilon_i, \Delta_i\} \longrightarrow H_{\mathrm{PMM}}(\theta). \tag{3.6}$$

During this conversion, fixed values are reinserted, homogeneous values are expanded across all sites or bonds when needed, and positive couplings such as t_i, U_i, and Δ_i are constrained to be non-negative by taking their absolute values. This separation between the optimization vector and the physical parameter dictionary makes it possible to use the same code for different ansätze.

In the main optimization runs, the hopping amplitudes are fixed to $t_i = 1$, while the interaction strength U is fixed to the value being studied. The remaining parameters, primarily the onsite energies ϵ_i and pairing amplitudes Δ_i, are optimized. For each system size n and interaction strength U, the lowest-loss physical parameter set is stored and later reused in the diagnostic and braiding calculations.

3.2.2 Parity classification inside the loss

For each trial Hamiltonian, the full many-body spectrum is obtained by exact diagonalization. The eigenstates are then classified by the total fermion parity operator,

$$P_{\text{tot}} = \prod_{i=1}^{n}(1 - 2n_i). \tag{3.7}$$

This gives two ordered energy lists, one in the even sector and one in the odd sector,

$$\{E_k^{(e)}\}, \qquad \{E_k^{(o)}\}. \tag{3.8}$$

The loss function compares these parity partners level by level. If the diagonalization produces an invalid parity structure, for example if one sector is missing or if the two sectors cannot be paired consistently, the trial point is discarded and the optimizer proceeds to another restart or candidate point.

3.2.3 Even-odd degeneracy term

The first loss contribution penalizes energy splitting between even- and odd-parity partners,

$$\delta_k = \left|E_k^{(e)} - E_k^{(o)}\right|. \tag{3.9}$$

These splittings are weighted more strongly at low energy, because the ground-state degeneracy is the most important Majorana signature. In the implementation, the weights decrease from a maximum value for the lowest parity pair to a smaller value for higher levels. The degeneracy contribution therefore has the schematic form

$$\mathcal{L}_{\text{deg}} = \sum_k w_k \left|E_k^{(e)} - E_k^{(o)}\right|, \qquad w_0 > w_1 > \cdots. \tag{3.10}$$

This term encourages the optimizer to produce an approximately parity-paired spectrum, with particular emphasis on the lowest-energy states.

3.2.4 Excitation-gap term

Degeneracy alone is not sufficient: the parity-paired states must also be separated from nearby levels. The second loss contribution penalizes small gaps within the parity-resolved spectra. For adjacent levels, the relevant gap is taken as the smaller of the even-sector and odd-sector spacings,

$$\Delta_k = \min\left(E_{k+1}^{(e)} - E_k^{(e)}, E_{k+1}^{(o)} - E_k^{(o)}\right). \tag{3.11}$$

Small gaps are penalized using a smooth threshold function,

$$\mathcal{L}_{\text{gap}} = \sum_k \tilde{w}_k \,\text{softplus}\left(\Delta_{\text{target}} - \Delta_k\right), \tag{3.12}$$

where Δ_{target} is the desired minimum gap scale. This term favors spectra where the relevant parity-paired states are not only close to each other, but also isolated from the rest of the spectrum.

3.2.5 Charge and Majorana-polarization terms

The loss also includes local diagnostics that test whether the parity partners resemble Majorana states rather than accidental degeneracies. The charge penalty measures how differently the even and odd partners occupy each site,

$$\mathcal{L}_Q = \sum_k \sum_j \left| \langle e_k | \, n_j \, | e_k \rangle - \langle o_k | \, n_j \, | o_k \rangle \right|. \tag{3.13}$$

A small value means that the parity partners have similar local charge distributions, consistent with non-local encoding.

The Majorana-polarization penalty compares the site-resolved Majorana-polarization profile with the expected edge-localized pattern. In the target configuration, the polarization is concentrated on the two ends of the chain with opposite sign, while the bulk sites carry little weight. Numerically, the penalty is evaluated from matrix elements of local Majorana operators between the even and odd parity partners. The sign of the target pattern is allowed to flip, since exchanging the labels of the two Majorana components changes the overall sign but not the physical localization pattern.

3.2.6 Total normalized loss

The full loss is a weighted sum of the spectral and local contributions. In the implementation, the degeneracy and gap terms are first collected into a single weighted spectral penalty array, after which the charge and Majorana-polarization penalties are added. Written in terms of the contributions above, the loss has the schematic form

$$\mathcal{L} = \frac{\mathcal{L}_{\text{deg}} + \mathcal{L}_{\text{gap}} + \mathcal{L}_Q + \mathcal{L}_{\text{MP}}}{\overline{|E|} + \eta}, \tag{3.14}$$

where $\overline{|E|}$ is the mean absolute energy scale of the spectrum and η is a small numerical offset. This normalization makes losses from different parameter sets more comparable and reduces the tendency to favor configurations only because the full spectrum has been rescaled.

3.2.7 Search strategy

The resulting loss landscape is non-convex, especially for interacting and inhomogeneous chains. The numerical search therefore combines local gradient-based optimization with repeated restarts and basin hopping. Local optimization is performed with Adam. At each iteration, the reduced vector θ is converted into physical parameters, the Hamiltonian is built and diagonalized, the loss is evaluated within the PyTorch optimization loop and used to update the free parameters with Adam.

Because a single local run can converge to a poor local minimum, the local optimization is repeated from randomly perturbed initial vectors. The lowest-loss result among these restarts is retained as the representative of that region of parameter space. Basin hopping is then used to explore different regions: a random displacement is applied to the current parameter vector, local optimization with restarts is performed again, and the new candidate is accepted if it improves the loss. Candidates with slightly worse loss can also be accepted with a Metropolis probability, allowing the search to escape shallow local minima.

For each pair (n, U), the best configuration found by this procedure is stored together with its loss, parameter configuration, and reconstructed physical parameters. These stored configurations form the input to the post-optimization diagnostics described in the next section.

3.3 Post-Optimization Diagnostics

The optimization procedure returns a scalar loss and a set of optimized parameters, but the loss alone is not sufficient to decide whether the result has a useful Majorana interpretation. After each optimization run, the best parameter set is therefore reconstructed and analyzed with a separate diagnostic workflow. The purpose of this step is to translate the numerical minimum into physical quantities that can be compared across system sizes, interaction strengths, and later braiding calculations.

3.3.1 Reconstruction of optimized Hamiltonians

The saved optimization output contains the raw optimization vector, the parameter configuration, the reconstructed physical parameters, and the final loss. For a chosen system size n and interaction strength U, the diagnostic routine selects the lowest-loss stored configuration. The raw vector is then converted back into the parameter dictionary described in Sec. 3.2, fixed values are reinserted, and the physical Hamiltonian

$$H_{\mathrm{PMM}}(\theta_{\mathrm{opt}}) \tag{3.15}$$

is rebuilt using the same Hamiltonian construction as in the optimization.

This reconstruction step is important because the diagnostics are not evaluated directly from the optimizer state. Instead, the optimized Hamiltonian is rebuilt from the stored physical parameters and diagonalized again. This makes the diagnostic plots and tables reproducible from the saved configuration files and ensures that the reported quantities correspond to the same Hamiltonian that is later used as input for the braiding calculations.

3.3.2 Parity-resolved spectrum

The reconstructed Hamiltonian is diagonalized exactly in the many-body Fock basis. The eigenstates are classified with the total fermion parity operator,

$$P_{\mathrm{tot}} = \prod_{i=1}^{n} (1 - 2n_i), \tag{3.16}$$

giving ordered even- and odd-parity spectra. For plotting and comparison, all energies are shifted so that the lowest state in either parity sector has energy zero. The shifted spectra make it possible to inspect whether the optimized configuration produces the expected near-degenerate parity pairs.

The most important scalar spectral diagnostic is the splitting of the lowest even-odd pair,

$$\delta_0 = \left| E_0^{(e)} - E_0^{(o)} \right|. \tag{3.17}$$

A small value of δ_0 indicates that the lowest even and odd states are approximately degenerate. However, this degeneracy is only useful if the pair is also separated from excited states. For this reason, the low-energy gap is defined as

$$\Delta_{\text{gap}} = \min\left(E_1^{(e)}, E_1^{(o)}\right) - \max\left(E_0^{(e)}, E_0^{(o)}\right), \tag{3.18}$$

after the common ground-state energy shift. This quantity measures the distance from the lowest parity pair to the nearest excited state in either parity sector. The combination of small δ_0 and large Δ_{gap} is the main spectral criterion used to identify promising optimized chains.

3.3.3 Local charge difference

Spectral degeneracy can occur for reasons unrelated to Majorana physics, so the post-optimization analysis also evaluates local observables in the even and odd states. The first local diagnostic is the site-resolved charge difference,

$$q_i = \left| \langle e_0 | \, n_i \, | e_0 \rangle - \langle o_0 | \, n_i \, | o_0 \rangle \right|. \tag{3.19}$$

The total charge difference reported in the results is

$$Q_0 = \sum_{i=1}^{n} q_i. \tag{3.20}$$

If the parity information is stored non-locally, the two parity partners should have nearly identical local charge distributions. A small Q_0 therefore indicates that local charge measurements cannot easily distinguish the even and odd states. In the diagnostic figures, the individual q_i values are plotted on top of the Majorana-polarization profile so that charge visibility and spatial localization can be compared site by site.

3.3.4 Majorana-polarization profile

The Majorana-polarization profile is computed from matrix elements of local Majorana operators between parity-partner states. In the summary diagnostics reported here, M_i denotes the profile of the lowest even-odd pair and measures where the parity-changing Majorana character is located. For an ideal finite PMM chain, the profile should be concentrated near the two ends, with opposite signs on the left and right edges and little weight in the bulk.

To summarize this profile in the results, two scalar quantities are extracted. The edge polarization is defined as the mean absolute polarization on the two end sites,

$$\overline{|M_{\text{edge}}|} = \frac{1}{2}\left(|M_1| + |M_n|\right), \tag{3.21}$$

while the bulk polarization is defined as the largest absolute value on the interior sites,

$$\max|M_{\text{bulk}}| = \max_{2 \le i \le n-1} |M_i|. \tag{3.22}$$

For two-dot systems there are no interior sites, so the bulk contribution is set to zero. A good Majorana-like configuration should have $\overline{|M_{\text{edge}}|}$ close to one and $\max|M_{\text{bulk}}|$ close to zero.

3.3.5 Selection for braiding calculations

The post-optimization diagnostics provide the bridge between the optimization results and the braiding part of the thesis. The parity-resolved spectra identify whether the optimized chain has a low-energy manifold suitable for projection. The charge and Majorana-polarization diagnostics then test whether this manifold has the expected non-local and edge-localized structure.

Although Majorana physics is often discussed in terms of the lowest parity pair, the optimization used here was deliberately applied to the full parity-resolved spectrum of the finite many-body Hamiltonian. This choice was motivated by the later braiding analysis, where projected spaces of increasing dimension, up to the full $2^9 = 512$ dimensional three-subsystem Hilbert space, are tested. The goal was therefore not only to obtain a clean ground-state Majorana signature, but also to find parameter sets whose parity pairing, excitation gaps, charge structure, and Majorana-polarization properties remain well behaved beyond the lowest states.

The post-optimization diagnostics therefore serve two purposes. First, they identify whether the lowest parity pair has the expected Majorana-like structure. Second, they indicate whether the optimized configuration remains suitable when larger projected subspaces are retained in the braiding calculations. Configurations that satisfy both requirements are selected as candidates for the effective Majorana construction described next.

3.4 Effective Majorana Operator Construction

The braiding calculations require explicit matrix representations of the Majorana operators associated with the optimized PMM chains. The theory chapter described why these operators should connect even and odd parity states. Here the focus is narrower: how the operators are constructed from the optimized eigenstates, embedded into the three-subsystem Hilbert space, projected to a chosen low-energy space, and checked before they are used in the braid.

3.4.1 Energy-level Majoranas from parity pairs

For each optimized subsystem, the Hamiltonian is diagonalized and the eigenstates are separated into even and odd parity sectors. The states are then paired by their order in energy. From the jth even-odd pair, two parity-changing operators are formed,

$$\gamma_{+,j} = |e_j\rangle \langle o_j| + |o_j\rangle \langle e_j| , \qquad \gamma_{-,j} = i \left(|e_j\rangle \langle o_j| - |o_j\rangle \langle e_j| \right) . \tag{3.23}$$

The effective Majorana operators used in the projected calculations are built by summing these contributions over a selected number of parity pairs,

$$\gamma_+^{(m)} = \sum_{j=0}^{m-1} \gamma_{+,j}, \qquad \gamma_-^{(m)} = \sum_{j=0}^{m-1} \gamma_{-,j}. \tag{3.24}$$

Here m is the number of even-odd pairs included in the operator construction. In the code this is controlled separately from the total projection dimension: the projection dimension determines how much of the full three-subsystem Hilbert space is retained, while m determines how many subsystem parity pairs are used to build the Majorana operator itself.

3.4.2 Embedding into the three-subsystem Hilbert space

The braiding geometry contains three optimized PMM subsystems, labelled A, B, and C. The full basis is ordered as described in Sec. 3.1, with all A sites first, followed by all B sites and then all C sites. After an operator has been constructed in the Hilbert space of one subsystem, it is embedded into the full Hilbert space using the same Jordan-Wigner ordering as the microscopic fermion operators.

For an operator γ_α acting on subsystem α, the embedding is schematically

$$\gamma_A \mapsto \gamma_A \otimes I_B \otimes I_C, \tag{3.25}$$

$$\gamma_B \mapsto JW_A \otimes \gamma_B \otimes I_C, \tag{3.26}$$

and

$$\gamma_C \mapsto JW_{AB} \otimes \gamma_C, \tag{3.27}$$

where JW_A is the Jordan-Wigner string over subsystem A, and JW_{AB} is the corresponding string over subsystems A and B. These strings are essential because the subsystem Majoranas must anticommute as fermionic operators after they are placed in the same Hilbert space.

3.4.3 Projection to the retained low-energy space

The full three-subsystem Hamiltonian is diagonalized, and a projection basis V is formed from the lowest d_P eigenvectors. Here d_P is the chosen projection dimension. Any full-space operator O is then represented in the retained space by

$$O_P = V^\dagger O V. \tag{3.28}$$

This projection is applied to the embedded Majorana operators before they are used in the projected braid models. The same projection basis is also used when projecting microscopic junction operators later, so that all operators act on the same retained low-energy Hilbert space.

3.4.4 Projected local components

In addition to the energy-level construction in Eq. 3.24, the code also constructs projected local Majorana components from the microscopic fermion operators on the relevant subsystem sites,

$$\Gamma_{+,P} = V^\dagger (c^\dagger + c) V, \qquad \Gamma_{-,P} = V^\dagger i(c^\dagger - c) V. \tag{3.29}$$

These projected local components provide a more microscopic reference for the Majorana operator at a chosen end of a subsystem. In the implementation, the energy-level operators are compared with these projected local components by minimizing a matrix-norm mismatch. When both components are allowed, the fitted operator has the form

$$\gamma_{\text{fit}} = a\,\gamma_+^{(m)} + b\,\gamma_-^{(m)}, \tag{3.30}$$

where the coefficients a and b are chosen from a least-squares fit in the projected operator space. This step is used to quantify how well the energy-level Majorana extracted from parity pairs matches the local microscopic Majorana component used in the projected-local braid comparison.

3.4.5 Normalization and algebra checks

After projection, the Majorana operators are no longer assumed to satisfy the ideal algebra exactly. The projected operators are therefore checked numerically before being used in the braid. The diagnostics test Hermiticity, normalization,

$$\gamma_i^2 \approx I, \tag{3.31}$$

and mutual anticommutation,

$$\{\gamma_i, \gamma_j\} \approx 0 \qquad (i \neq j). \tag{3.32}$$

The matrix mismatch between the fitted energy-level operator and the projected local component is also stored. These checks are important because the later braid errors should not be interpreted only as failures of the pulse protocol: they can also reflect imperfect Majorana operators after projection.

3.5 Braiding Models and Pulse Protocol

The braiding calculations use the optimized PMM chains as building blocks for a three-subsystem junction. The purpose of this section is to define the Hamiltonians that are driven during the braid and the common pulse sequence used for all model variants. The following section then describes how the resulting time-dependent evolution is transported and compared with the target exchange gate.

3.5.1 Three-subsystem geometry

The junction consists of three optimized PMM subsystems, labelled A, B, and C. Subsystem A acts as the reference arm and supplies two Majorana operators, denoted γ_0 and γ_1. The Majorana operators to be exchanged are associated with subsystems B and C, and are denoted γ_2 and γ_3. In this notation, the intended exchange is generated by moving the coupling of γ_0 from γ_1 to γ_2, then to γ_3, and finally back to γ_1.

All braiding models are built in the projected Hilbert space introduced in Sec. 3.4. The projection dimension determines how many low-energy states of the full three-subsystem Hamiltonian are retained. Within this common projected space, different choices of coupling operators can be compared directly.

In the projection scans, the driven braid terms can also be supplemented by the projected static Hamiltonian of the uncoupled subsystem background,

$$H_{\text{static}} = V^\dagger \left(H_B + H_C\right) V. \tag{3.33}$$

This term keeps the internal optimized-chain structure of subsystems B and C while the time-dependent junction couplings are varied. Since it is included in the same way for the model comparisons, the differences between the braid results come from the choice of driven coupling operators rather than from a different static background.

3.5.2 Ideal four-Majorana model

The simplest braid model uses only four Majorana operators and hand-chosen bilinear couplings. In the projected space, the time-dependent Hamiltonian is

$$H_{\text{ideal}}(t) = \lambda_1(t)\, i\gamma_0\gamma_1 + \lambda_2(t)\, i\gamma_0\gamma_2 + \lambda_3(t)\, i\gamma_0\gamma_3. \tag{3.34}$$

This model is used as a reference calculation. Since the couplings are written directly as Majorana bilinears, it isolates the braid protocol itself from the additional complications of microscopic junction operators and imperfect projected Majoranas.

3.5.3 Projected microscopic junction model

To connect the braid more directly to the optimized dot-chain Hamiltonian, the ideal bilinears can be replaced by microscopic junction operators. The junction between subsystems A and B is constructed from ordinary hopping and pairing terms between the endpoint dots,

$$H_{AB}^{\mathrm{junc}} = -t_{\mathrm{j}}\left(c_A^\dagger c_B + c_B^\dagger c_A\right) + \Delta_{\mathrm{j}}\left(c_A c_B + c_B^\dagger c_A^\dagger\right), \tag{3.35}$$

with an analogous operator H_{AC}^{junc} for the A–C junction. These full-space junction operators are projected into the retained low-energy space,

$$T_{AB} = V^\dagger H_{AB}^{\mathrm{junc}} V, \qquad T_{AC} = V^\dagger H_{AC}^{\mathrm{junc}} V. \tag{3.36}$$

The corresponding projected microscopic braid Hamiltonian is then

$$H_{\mathrm{junc}}(t) = \lambda_1(t)\, T_A + \lambda_2(t)\, T_{AB} + \lambda_3(t)\, T_{AC}, \tag{3.37}$$

where $T_A = i\gamma_0\gamma_1$ fixes the initial pairing inside subsystem A. This model tests whether the physical junction couplings reproduce the desired Majorana exchange after projection.

3.5.4 Projected local-operator model

A second microscopic comparison keeps the bilinear braid structure but replaces the ideal exchanged Majoranas by projected local Majorana components. For example, on the relevant endpoint of subsystem B one constructs

$$\Gamma_{B,-} = V^\dagger i(c_B^\dagger - c_B)V, \tag{3.38}$$

and similarly for subsystem C. After normalization, these local projected operators define $\gamma_{2,\mathrm{loc}}$ and $\gamma_{3,\mathrm{loc}}$. The local-operator braid Hamiltonian is

$$H_{\mathrm{loc}}(t) = \lambda_1(t)\, i\gamma_0\gamma_1 + \lambda_2(t)\, i\gamma_0\gamma_{2,\mathrm{loc}} + \lambda_3(t)\, i\gamma_0\gamma_{3,\mathrm{loc}}. \tag{3.39}$$

This model is useful because it sits between the ideal and fully microscopic descriptions. The projected microscopic model first projects the full junction operator, while the projected local-operator model first projects local Majorana components and then forms bilinears from them. It therefore keeps a clear Majorana-bilinear form, but the operators being coupled are obtained from projected local fermionic operators rather than only from energy-level parity pairs.

3.5.5 Pulse sequence

The same pulse sequence is used for each braid model. To avoid confusing the braid amplitudes with the superconducting pairing parameters Δ_i, the time-dependent braid couplings are denoted by $\lambda_i(t)$. The coupling envelope is a smooth pulse constructed from two sigmoid functions,

$$\lambda(t;T_p) = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \frac{1}{1+\exp\left[-s(t-T_p+w/2)\right]} \frac{1}{1+\exp\left[s(t-T_p-w/2)\right]}, \tag{3.40}$$

where T_p is the pulse center, w is the pulse width, and s controls the steepness of the rise and fall. The three braid couplings are then chosen as

$$\lambda_1(t) = \lambda(t;0) + \lambda(t;T) - \lambda_{\min}, \qquad \lambda_2(t) = \lambda(t;T/3), \qquad \lambda_3(t) = \lambda(t;2T/3). \tag{3.41}$$

Thus the A-arm coupling is active at the beginning and end of the cycle, while the A–B and A–C couplings are activated in sequence during the middle of the protocol. In the numerical calculations used here, the total braid time is set to $T = 1$, with $\lambda_{\max} = 1$, $\lambda_{\min} = 0$, pulse width $w = T/3$, and steepness $s = 20/w$.

At each time step, the Hamiltonian for the chosen model is assembled from the same three time-dependent coefficients. The instantaneous spectrum is monitored along the path, and the isolated low-energy band is then transported using the Kato procedure described in the next section.

3.6 Kato Transport and Braid Validation

After a braid Hamiltonian has been defined, the remaining task is to compute the unitary generated by the adiabatic path and to check whether it implements the desired Majorana exchange. The calculation is performed in the projected Hilbert space, but the transported subspace can be smaller than the full projection space. This section describes how that transported subspace is selected, how the Kato evolution is computed, and which diagnostics are used to compare the resulting unitary with the target braid.

3.6.1 Choice of transported subspace

For a given projected Hamiltonian $H(t)$, the instantaneous spectrum is sampled along the braid path. Candidate transported dimensions are tested by looking at the minimum spectral gap above each candidate band throughout the protocol. If the transported dimension is d_T, the relevant gap is

$$\Delta_T = \min_t \left[E_{d_T}(t) - E_{d_T-1}(t)\right], \tag{3.42}$$

where the eigenvalues are ordered increasingly and the index convention starts at zero. The numerical procedure selects the candidate d_T with the largest persistent gap. This choice avoids fixing the transported manifold by hand when the projection dimension is increased, and instead follows the most clearly isolated low-energy band available in the chosen projected model.

3.6.2 Numerical Kato transport

The braid path is discretized into time points $t_0, \dots, t_N$. At each time step, the Hamiltonian is diagonalized and the projector onto the transported band is constructed,

$$P(t_\ell) = \sum_{a=1}^{d_T} |\psi_a(t_\ell)\rangle \langle\psi_a(t_\ell)|. \tag{3.43}$$

The Kato generator is approximated from consecutive projectors,

$$K(t_\ell) = P(t_\ell)\dot{P}(t_\ell) - \dot{P}(t_\ell)P(t_\ell), \qquad \dot{P}(t_\ell) \approx \frac{P(t_{\ell+1}) - P(t_\ell)}{\Delta t}. \tag{3.44}$$

The transport unitary is updated as

$$U_K(t_{\ell+1}) = \exp[-\Delta t\, K(t_\ell)]U_K(t_\ell), \qquad U_K(t_0) = I. \tag{3.45}$$

This convention matches the numerical implementation and gives the geometric transport of the selected subspace along the braid path. The instantaneous energies and pulse amplitudes are stored at the same time, so that the quality of the transported band can be checked together with the final braid unitary.

3.6.3 Target exchange gate

The target unitary for exchanging γ_2 and γ_3 is taken to be

$$U_{\text{target}} = \exp\left(-\frac{\pi}{4}\gamma_2\gamma_3\right). \tag{3.46}$$

This convention corresponds to the exchange action

$$\gamma_2 \mapsto -\gamma_3, \qquad \gamma_3 \mapsto \gamma_2, \tag{3.47}$$

up to the overall sign convention chosen for the Majorana operators. Since a global phase of the transported unitary is physically irrelevant, comparisons with the target gate are made after aligning the overall phase. In practice, the reported gate error is a Frobenius-norm distance between the transported unitary and the phase-aligned target unitary, restricted to the initial transported basis.

3.6.4 Operator-action checks

The first validation step checks the induced action on the Majorana operators directly. For each model, the transported unitary is applied as

$$\gamma_i' = U_K^\dagger \gamma_i U_K. \tag{3.48}$$

The resulting operators are compared with the expected single-exchange map,

$$\gamma_2' \approx -\gamma_3, \qquad \gamma_3' \approx \gamma_2, \qquad \gamma_0' \approx \gamma_0, \qquad \gamma_1' \approx \gamma_1. \tag{3.49}$$

The mismatch is measured using a normalized Frobenius norm,

$$\|A\|_{\text{norm}} = \frac{\|A\|_F}{\sqrt{d}}, \tag{3.50}$$

where d is the dimension of the matrix being compared. These checks are useful because they test the braid as an operator transformation, rather than only as a matrix acting on a chosen basis.

3.6.5 Parity-resolved gate checks

The transported unitary is also analyzed in a parity basis. The total parity operator is projected into the initial transported space and diagonalized, separating the unitary into odd- and even-parity blocks. The off-block leakage is defined from the norm of the odd-to-even and even-to-odd blocks,

$$\mathcal{L}_{\text{parity}} = \|U_{oe}\| + \|U_{eo}\|. \tag{3.51}$$

A small value indicates that the braid approximately preserves fermion parity inside the transported manifold.

After the parity blocks have been separated, each block is compared with the corresponding block of the target exchange gate in Eq. 3.46. This produces separate odd- and even-sector target-gate errors. Reporting the parity blocks separately is useful because the same Majorana exchange can appear with different matrix representations in the two parity sectors, while still describing the same physical braid.

3.6.6 Energy-level and projected-local targets

The final comparison distinguishes between two possible choices of target Majoranas. The energy-level target uses the Majorana operators constructed from even-odd parity pairs, as described in Sec. 3.4. The projected-local target instead uses the normalized projected local operators obtained from microscopic combinations such as $i(c^\dagger - c)$. These two targets need not be identical after projection, especially when more excited states are retained or when the optimized Majorana operators are not perfectly local.

For this reason, the physical braid is compared in several ways: against the energy-level target, against the projected-local target, and against the ideal reference braid. The mismatch between the two target constructions is also recorded. This separation is important for interpreting the results. If the physical braid disagrees with the energy-level target but agrees with the projected-local target, the issue may be the identification of the effective Majorana operators rather than the adiabatic pulse itself. Conversely, if both target comparisons fail, the projected braid Hamiltonian is not implementing the desired exchange within the chosen transported subspace.

Chapter 4

Results

This chapter presents the main numerical results of the thesis. The first part analyzes the optimized PMM chains obtained from the basin-hopping optimization procedure. The optimized parameters are compared across system sizes and interaction strengths, and the resulting Hamiltonians are evaluated using the spectral and local Majorana diagnostics introduced in the previous chapters.

The second part focuses on braiding. Based on the post-optimization diagnostics, the most promising optimized configuration is selected as the building block for a three-subsystem braiding geometry. The braid analysis begins with a controlled low-energy comparison between the ideal four-Majorana model, the projected microscopic junction model, and the projected local-operator model. These calculations serve as benchmarks for whether the optimized microscopic system reproduces the exchange behavior expected from the effective Majorana description.

The final part extends the braid analysis to larger projected Hilbert spaces, up to the full $2^9 = 512$ dimensional three-subsystem system. This tests how robust the low-energy agreement remains when higher sectors are retained, and clarifies the distinction between energy-level Majorana constructions and physically projected local-operator constructions.

4.1 Optimized PMM Chains

This section summarizes the parameter configurations obtained from the basin-hopping optimization procedure and compares how they depend on chain length and interaction strength. The optimization was performed for chain lengths $n = 2, 3$, and 4, and for fixed interaction strengths $U = 0, 0.1, 0.5, 1.0, 2.0$, and 5.0. In all cases, the hopping amplitudes were fixed to $t_i = 1$, while the onsite energies ϵ_i and pairing amplitudes Δ_i were optimized.

Fixing the hopping scale makes the optimized parameters easier to compare across system sizes and interaction strengths. It also prevents the optimizer from reducing the relative importance of interactions through a trivial global rescaling of the hopping and pairing amplitudes. The purpose of this section is therefore to identify the lowest-loss parameter sets and to extract the main trends in the optimized onsite and pairing profiles before applying the post-optimization diagnostics.

4.1.1 Lowest-loss configurations

Table 4.1 summarizes the lowest-loss configuration found for each combination of chain length n and fixed interaction strength U. Since the hopping amplitudes were fixed to $t_i = 1$ in these runs, the table lists only the optimized pairing amplitudes and onsite energies. Empty entries correspond to parameters that are absent for shorter chains.

n	U	Loss	Δ_1	Δ_2	Δ_3	ϵ_1	ϵ_2	ϵ_3	ϵ_4
2	0.0	5.657e-03	1.000	–	–	-0.001	0.002	–	–
2	0.1	2.617e-01	1.047	–	–	-0.050	-0.047	–	–
2	0.5	1.083e+00	1.249	–	–	-0.255	-0.249	–	–
2	1.0	1.940e+00	1.498	–	–	-0.499	-0.496	–	–
2	2.0	3.203e+00	2.000	–	–	-1.006	-1.004	–	–
2	5.0	4.004e+00	3.499	–	–	-2.508	-2.506	–	–
3	0.0	3.442e-01	1.010	1.348	–	0.019	1.774	-0.466	–
3	0.1	5.603e-01	1.001	1.002	–	-0.101	-0.607	-0.003	–
3	0.5	9.359e-01	1.004	1.012	–	-0.046	-0.889	-0.454	–
3	1.0	1.452e+00	1.025	1.130	–	-0.195	-1.530	-0.802	–
3	2.0	2.326e+00	1.584	0.978	–	-1.511	-3.077	-0.484	–
3	5.0	3.142e+00	0.846	1.746	–	-4.814	1.009	-0.184	–
4	0.0	3.161e-01	2.698	5.332	1.004	-0.059	-4.252	0.236	-0.003
4	0.1	4.840e-01	1.790	5.002	1.013	0.156	2.376	-1.966	-0.050
4	0.5	1.006e+00	1.212	3.490	1.206	-0.246	-0.499	-0.503	-0.250
4	1.0	5.670e-01	2.238	18.807	1.072	-0.500	-1.028	-1.025	-0.496
4	2.0	1.824e+00	1.492	5.537	1.588	-0.869	3.526	-6.508	-1.220
4	5.0	2.600e+00	3.389	9.513	3.336	-2.645	5.032	-15.431	-2.293

Table 4.1: Lowest-loss optimized configurations extracted from the optimization procedure for each combination of system size n and fixed interaction strength U.

The two-dot results provide a useful reference point. In the non-interacting case, the optimizer reproduces the expected sweet spot, with $\Delta_1 \approx t_1 = 1$ and onsite energies close to zero. As the interaction strength is increased, the optimized pairing amplitude grows above the fixed hopping scale, while the onsite energies move approximately toward $\epsilon_i \approx -U/2$. This indicates that the optimizer deforms the known two-dot solution in a smooth and physically interpretable way.

At the same time, the loss increases substantially with interaction strength. This suggests that the interacting two-dot system becomes progressively less compatible with the full set of optimization criteria, even though the optimized parameters continue to follow the expected interacting trend. The table therefore motivates the more detailed spectral and local diagnostics presented below, rather than by itself establishing whether the optimized states have useful Majorana character.

For the three-dot system, the weak- and intermediate-interaction configurations remain close to the homogeneous pairing limit, with $\Delta_1 \approx \Delta_2 \approx 1$ for $U = 0.1$ and $U = 0.5$. The onsite energies, however, become inhomogeneous, with the middle dot often shifted more strongly than the edge dots. This suggests that the optimizer uses the additional site to reshape the effective potential profile while keeping the pairing amplitudes near the hopping scale.

The four-dot results show a more complex optimization landscape. Several of the lowest-loss configurations have strongly enhanced central pairing amplitudes, while the outer pairing amplitudes remain closer to unity. The onsite energies also become increasingly inhomogeneous, especially at larger interaction strengths. These trends indicate that longer PMM chains do not simply reproduce the two-dot sweet spot, but instead favor structured parameter profiles that depend sensitively on both n and U.

One recurring feature in the longer chains is that interior onsite energies are often shifted more strongly than the edge onsite energies. A possible interpretation is that the optimizer is creating an effective barrier in the interior of the chain, helping separate the Majorana-like modes toward the edges. This interpretation is tested more directly in the following sections through the parity-resolved spectra, charge diagnostics, and Majorana-polarization profiles.

4.1.2 Representative optimized spectra

The representative spectrums in Figs. 4.1 and 4.2 illustrate how interactions affect the optimized two-dot system. In the non-interacting case, both parity pairs are nearly degenerate, consistent with the expected two-dot sweet spot. When a weak interaction is introduced at $U = 0.1$, the lowest parity pair remains close to degenerate, while the excited pair begins to show a visible splitting. At $U = 2.0$, this excited-state splitting becomes much larger. Meaning that although the ground-state pair can retain a low-energy Majorana-like signature, it still does not maintain clean parity-pair degeneracy across the full finite spectrum.

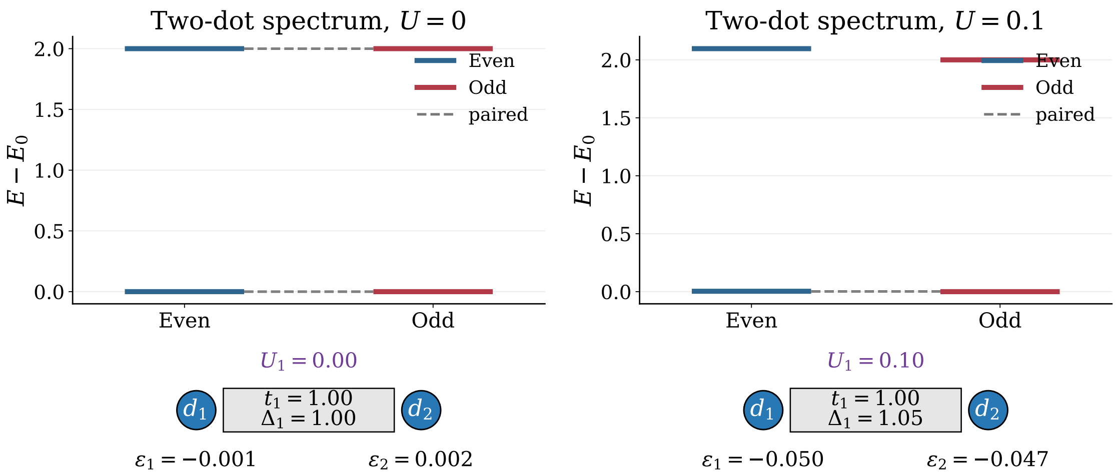

Figure 4.1: Parity-resolved spectra after optimization for the two-dot system at $U = 0$ and $U = 0.1$. Energies are shifted by the ground-state energy in each panel, and the optimized parameters are shown schematically below each spectrum.

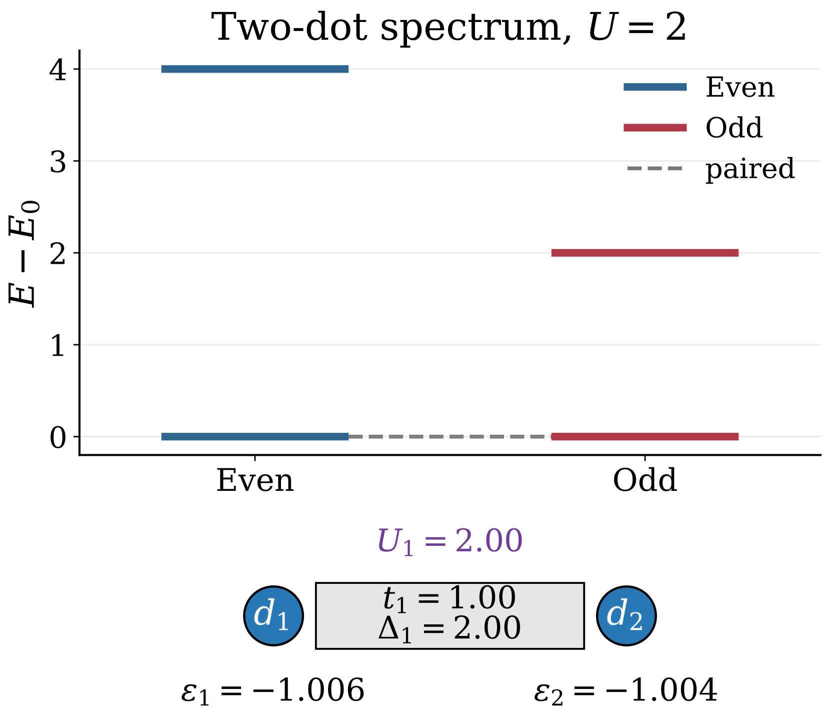

Figure 4.2: Parity-resolved spectrum after optimization for the two-dot system at $U = 2.0$. Energies are shifted by the ground-state energy, and the optimized parameters are shown schematically below the spectrum.

The optimized three-dot system at $U = 0.1$ provides a more promising representative configuration. As shown in Fig. 4.3, the parity-resolved spectrum remains nearly paired, while the Majorana-polarization profile has the expected edge-localized structure. The polarization is large and opposite in sign on the two outer dots, and it is strongly suppressed on the middle dot. This combination of spectral pairing and spatial Majorana character motivates using this configuration as the main candidate for the braiding calculations below. Comparing the three-dot configuration for $U = 0.1$ with the two-dot configuration for $U = 0.1$ also illustrates how the additional site can help stabilize the Majorana-like features in the presence of interactions, by noticing the excited-states retaining a better energy degeneracy in the three-dot case than in the two-dot case, even though both configurations have similar ground-state splitting and similar optimized parameters. The additional site therefore provides more freedom to optimize the excited states, which is important for maintaining a clean low-energy manifold that can support braiding.

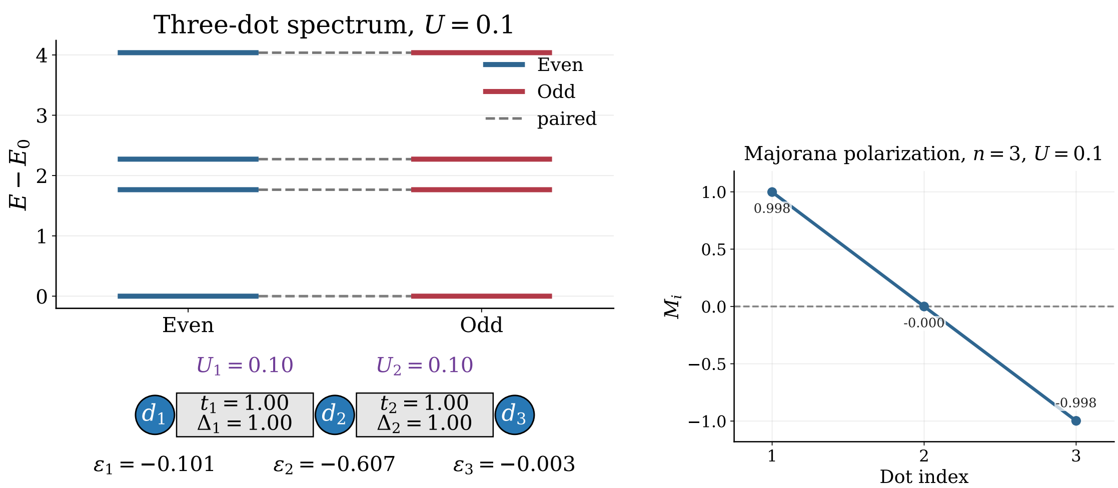

Figure 4.3: Parity-resolved spectrum and Majorana-polarization profile for the optimized three-dot system at $U = 0.1$. The left panel shows the optimized spectrum and parameters, while the right panel shows the Majorana polarization of the lowest even-odd pair.

4.2 Post-Optimization Diagnostics

The optimized parameter values alone do not determine whether a configuration is useful for Majorana braiding. After optimization, the candidate Hamiltonians must therefore be tested through spectral and local diagnostics. The parity-resolved spectra and low-energy gaps show whether the optimizer has produced an isolated nearly degenerate low-energy manifold, while the charge difference and Majorana-polarization profile test whether this manifold has the expected non-local and edge-localized Majorana character.

4.2.1 Parity-resolved spectra and low-energy gaps

Figure 4.4 compares the parity-resolved spectra for $n = 2, 3$, and 4 at selected interaction strengths. A useful Majorana-like configuration should have a nearly degenerate lowest even-odd pair, together with a finite gap separating this pair from higher excited states. The dashed connectors in the figure indicate parity pairs that remain close in energy after optimization.

The non-interacting cases give the cleanest spectra, as expected. For the two-dot system, the parity pairing is nearly ideal at $U = 0.0$, but the excited-state splitting becomes visible already at weak interaction and increases further at stronger interaction. This shows that the interacting two-dot system can retain a low-energy degeneracy while losing clean parity pairing across the full finite spectrum.

The three-dot system is more stable across all interaction strengths shown, compared to both the two-dot and four-dot systems. For all three interaction strengths, $U = 0.0$, $U = 0.1$, and $U = 2.0$, we can see that all energy levels are degenerate in pairs, up to a numerical threshold, with a decently large gap separating the energy levels into distinct energy sectors. The four-dot system, shows a promising structure for weak and no interactions, while we can clearly see that for stronger interactions $U = 2.0$, the energy degeneracy falls apart. Most importantly, the ground state energies are no longer degenerate, and thus there is no longer a clear low-energy manifold that can support braiding. The three-dot system therefore appears to be the most robust candidate for maintaining a clean low-energy manifold across different interaction strengths, but

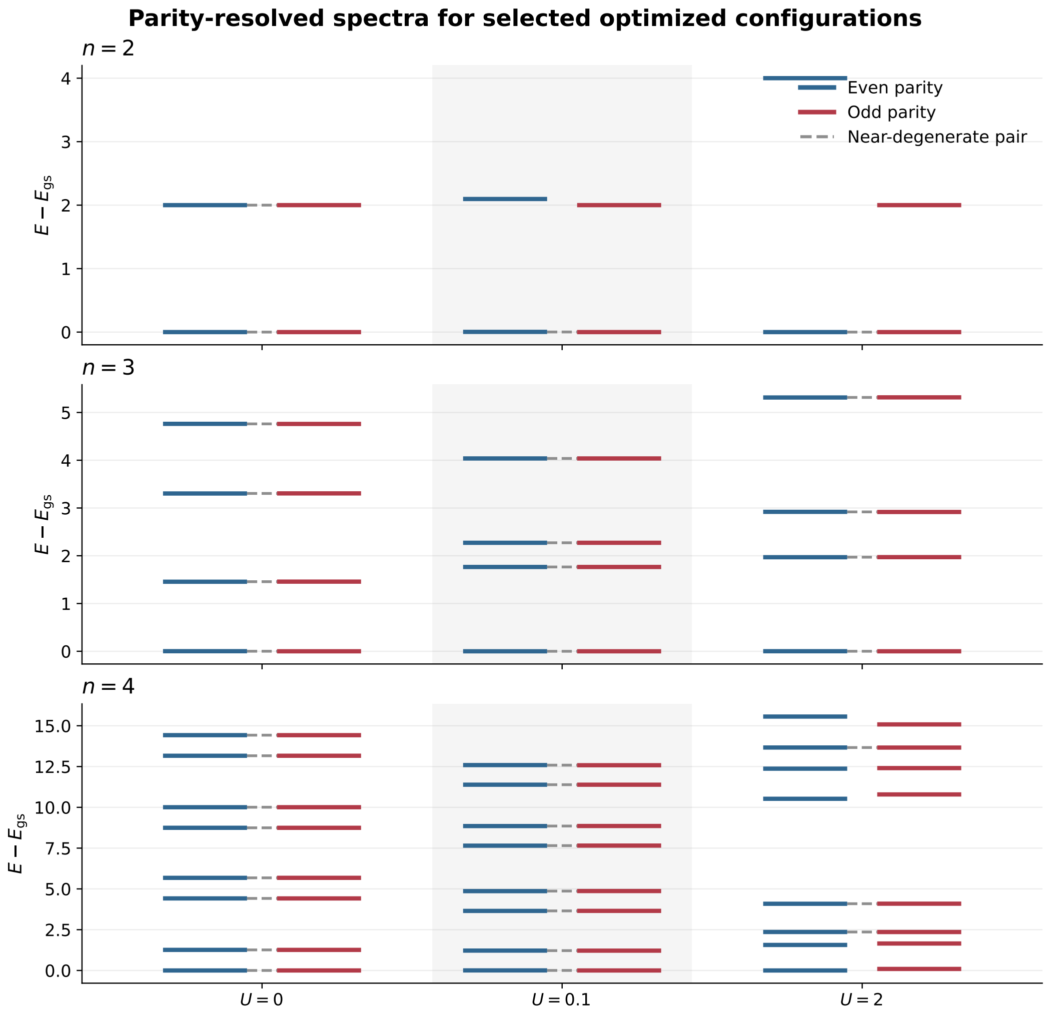

Figure 4.4: Parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size n, and energies are shifted so that the ground state is at zero.

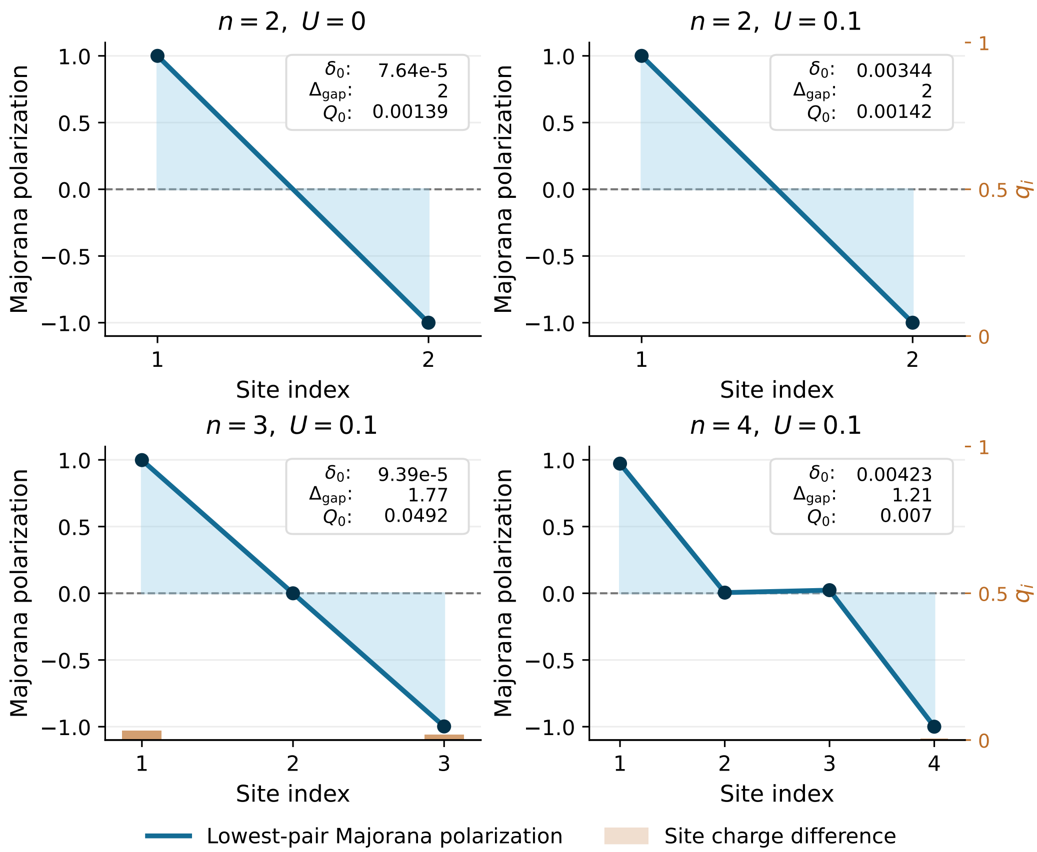

Figure 4.5: Local diagnostics for representative optimized configurations. The blue curve shows the Majorana-polarization profile of the lowest even-odd pair, while the orange bars, read against the 0–1 right axis, show the local charge difference $q_i = |\langle e|n_i|e\rangle - \langle o|n_i|o\rangle|$ on each site. The inset in each panel reports the ground-state splitting δ_0, the low-energy gap Δ_{gap}, and the total charge difference Q_0.

in order to determine if it is possible to braid using the three-dot system in a weak interaction regime, we need to also check the local diagnostics and Majorana character.

4.2.2 Local diagnostics and Majorana character

As discussed in the methodology chapter, Majorana-like states cannot be identified from spectral degeneracy alone. The parity-resolved spectra and low-energy gaps test whether the optimized Hamiltonian contains an isolated, nearly degenerate low-energy manifold. The present subsection examines the complementary local diagnostics: the Majorana-polarization profile and the local charge difference between the lowest even-odd parity partners.

A convincing Majorana-like configuration should have opposite Majorana polarization on the two ends of the chain, with strongly suppressed polarization in the interior, as quantified by Eqs. 3.21 and 3.22. In addition, the even and odd parity partners should be locally difficult to distinguish. This is tested by the charge-difference diagnostic in Eq. 3.20; smaller values indicate that the parity information is less visible to local charge measurements.

n	U	δ_0	Δ_{gap}	Q_0	$\overline{\lvert M_{\text{edge}}\rvert}$	$\max\lvert M_{\text{bulk}}\rvert$
2	0	7.64e-05	2	0.00139	1	0
2	0.1	0.00344	2	0.00142	1	0
3	0.1	9.39e-05	1.77	0.0492	0.998	0.000202
4	0.1	0.00423	1.21	0.007	0.986	0.0233

Table 4.2: Summary of local diagnostics for the representative configurations shown in Fig. 4.5. Here δ_0 is the splitting of the lowest even-odd pair, Δ_{gap} is the gap separating this pair from the next excited states, Q_0 is the total charge difference between the parity partners, $\overline{|M_{\text{edge}}|}$ is the mean absolute Majorana polarization on the first and last sites, and $\max|M_{\text{bulk}}|$ is the largest absolute polarization on the interior sites.

Figure 4.5 compares the spatial profiles directly, while Table 4.2 summarizes the corresponding scalar diagnostics. For the two-dot system, the edge polarization is essentially ideal, with $\overline{|M_{\text{edge}}|} \approx 1$ and no bulk contribution by construction. The total charge difference is also very small, $Q_0 \approx 1.4 \times 10^{-3}$, both at $U = 0$ and at $U = 0.1$. These two-dot cases are useful references, but because they have no interior sites they do not test suppression of Majorana weight in the bulk.

The three-dot configuration at $U = 0.1$ gives the clearest interacting Majorana-polarization pattern. The ground-state splitting is only $\delta_0 = 9.39 \times 10^{-5}$, the excitation gap remains large, $\Delta_{\text{gap}} = 1.77$, and the bulk polarization is almost completely suppressed, with $\max|M_{\text{bulk}}| = 2.02 \times 10^{-4}$. Its charge difference, however, is larger than in the two-dot and four-dot reference cases, with $Q_0 = 0.0492$. This shows that the three-dot state is not ideal in every diagnostic, even though its spectral and polarization properties are the strongest.

The four-dot configuration at $U = 0.1$ shows the opposite tradeoff. It has a smaller charge difference, $Q_0 = 0.0070$, and comparably strong edge polarization, $\overline{|M_{\text{edge}}|} \approx 0.986$. However, the ground-state splitting increases to $\delta_0 = 4.23 \times 10^{-3}$ and the bulk polarization rises to $\max|M_{\text{bulk}}| = 2.33 \times 10^{-2}$.

Taken together, these diagnostics show that different system sizes optimize different aspects of the Majorana criteria. The four-dot case performs better in terms of local charge indistinguishability, while the three-dot case performs better in terms of spectral near-degeneracy and suppression of bulk Majorana weight. Since the braiding analysis requires a low-energy subspace that is both nearly degenerate and spatially well localized, the interacting three-dot system remains the most convincing candidate for the rest of the thesis.

4.3 Selection of Braiding Candidate

The braiding calculations are performed using the optimized three-dot configuration at $U = 0.1$. This choice is based on the combined post-optimization diagnostics above. Although the four-dot configuration has a smaller local charge difference, the three-dot configuration has the most favorable combination of small ground-state splitting, large excitation gap, clear edge-localized Majorana-polarization profile, and strongly suppressed bulk polarization. Since the braiding analysis relies on a nearly degenerate and spatially localized low-energy subspace, the three-dot system is chosen as the main microscopic building block.

The optimized parameters used for each three-dot subsystem are

$$\Delta_1 = 1.001, \qquad \Delta_2 = 1.002, \qquad \epsilon_1 = -0.101, \qquad \epsilon_2 = -0.607, \qquad \epsilon_3 = -0.003, \tag{4.1}$$

with fixed hopping amplitudes $t_i = 1$ and fixed interaction strength $U = 0.1$.

In the braiding geometry, three identical copies of this optimized three-dot chain are combined. Each subsystem has Hilbert-space dimension $2^3 = 8$, giving a total microscopic Hilbert space of dimension $2^9 = 512$. This system is large enough to test braiding beyond the lowest projected manifold, while still being small enough for exact diagonalization and systematic projection scans. The selected configuration therefore provides a useful setting for studying how interactions and larger projection dimensions affect the braiding behavior.

4.4 Braiding in Projected Majorana Models

The optimized three-dot chain is now used as the microscopic building block for the braiding calculations. Before studying larger projection dimensions, I first establish a controlled low-energy benchmark by comparing three related braid models: the ideal four-Majorana reference model, the projected microscopic junction model, and the projected local-operator model. This comparison tests whether the optimized microscopic system reproduces the exchange behavior expected from the effective Majorana description.

In the ideal reference model, the braid is generated directly by tunable Majorana bilinears built from the four Majorana operators defined in Eq. 2.15. This isolates the braid protocol itself from microscopic details. In the projected microscopic model, these ideal bilinears are replaced by projected junction operators constructed from ordinary hopping and crossed-Andreev pairing terms between endpoint dots. For example, the A–B junction is

$$H_{AB}^{\text{junc}} = -t_{\text{j}} \left(c_A^\dagger c_B + c_B^\dagger c_A \right) + \Delta_{\text{j}} \left(c_A c_B + c_B^\dagger c_A^\dagger \right), \tag{4.2}$$

as introduced in Eq. 3.35. After projection into the retained low-energy subspace, this operator gives the effective coupling T_{AB} used in the driven braid Hamiltonian of Eq. 3.37. The projected microscopic braid therefore tests whether physically motivated junction couplings reproduce the same exchange channel as the ideal Majorana model.

4.4.1 Ideal four-Majorana reference

Figure 4.6 shows the reference braid generated by four active Majorana operators in the projected low-energy space. The three couplings are turned on sequentially, so that the dominant coupling is first between γ_0 and γ_1, then between γ_0 and γ_2, and finally between γ_0 and γ_3 before returning to the initial configuration. Throughout this cycle, the transported low-energy manifold remains essentially four-fold degenerate. The largest instantaneous splitting within this manifold is only $\max(E_3 - E_0) = 2.28 \times 10^{-12}$, while the minimum gap to the excited states remains open at $\min(E_4 - E_3) = 1.414$.

Using Kato transport, the resulting unitary reproduces the expected single-exchange action on the Majorana operators. The largest error in the transformations $\gamma_2 \to -\gamma_3$ and $\gamma_3 \to \gamma_2$ is 1.33×10^{-5}. In the parity-resolved gate blocks, the odd and even 2×2 blocks agree with the target exchange gate to within 6.53×10^{-8}, while the off-block leakage is only 1.02×10^{-11}. The effective model therefore provides a clean benchmark for the more microscopic projected calculations.

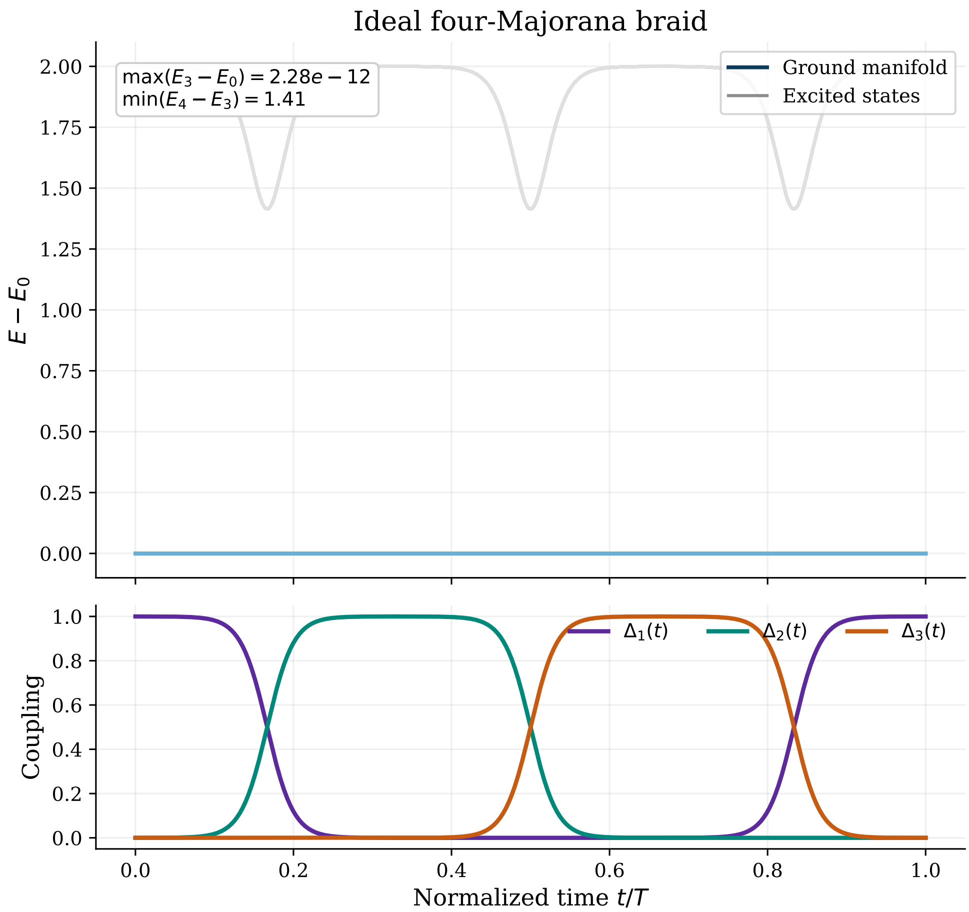

Figure 4.6: Idealized four-Majorana braiding protocol. The top panel shows the instantaneous spectrum of the effective Hamiltonian during the braid, with all energies shifted so that the lowest state remains at zero. The bottom panel shows the three smooth pulse couplings that move the dominant Majorana coupling through the network.

4.4.2 Projected microscopic and local-operator braids

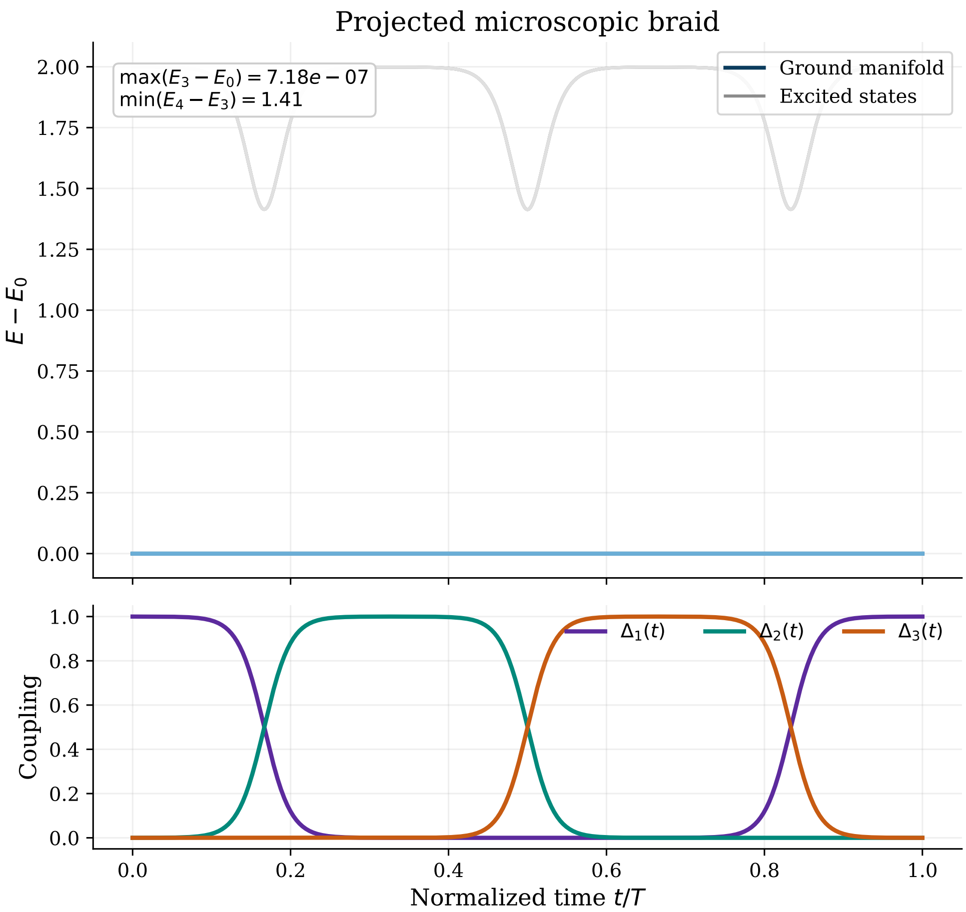

Figure 4.7: Braiding protocol after replacing the idealized couplings by projected microscopic junction operators. The spectral evolution remains almost identical to the effective-model benchmark, with a well-isolated low-energy manifold throughout the cycle.

After replacing the ideal Majorana-bilinear couplings by projected microscopic junction operators, the braid remains close to the reference result. Figure 4.7 shows that the four-dimensional low-energy manifold is still well separated from higher states throughout the cycle. The maximum splitting inside this manifold increases only to $\max(E_3 - E_0) = 7.18 \times 10^{-7}$, while the minimum gap remains large, $\min(E_4 - E_3) = 1.413$.

The junction decomposition in Fig. 4.8 explains why the projected microscopic braid agrees so well with the ideal reference. The projected AB coupling is dominated by the bilinear $i\gamma_{A1}\gamma_{B2}$ with coefficient -0.999, and the projected AC coupling is dominated by $i\gamma_{A1}\gamma_{C2}$ with approximately the same coefficient. This identifies γ_{B2} and γ_{C2} as the Majoranas that participate in the microscopic braid, while γ_{A1} acts as the shared Majorana through which the exchange is mediated.

As an additional consistency check, I also compare with a projected local-operator braid. In this construction, the exchanged Majorana channels are obtained by projecting local endpoint Majorana operators rather than by projecting the full microscopic junction Hamiltonian. This tests whether the same low-energy exchange channel is accessed by local microscopic operators.

Table 4.3 shows that the microscopic braid reproduces the ideal benchmark extremely

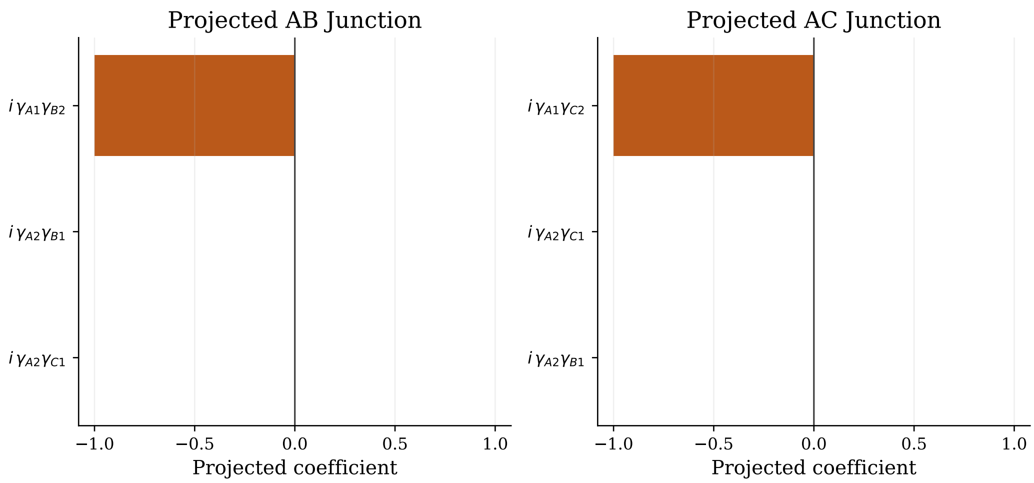

Figure 4.8: Dominant Majorana-bilinear components of the projected microscopic junction operators. The AB junction is almost entirely $i\gamma_{A1}\gamma_{B2}$, while the AC junction is almost entirely $i\gamma_{A1}\gamma_{C2}$. All subleading bilinears are strongly suppressed.

Model	$\max(E_3 - E_0)$	$\min(E_4 - E_3)$	Max 1x err.	Leakage	Odd err.	Even err.
Ideal effective model	2.275e-12	1.414	1.334e-05	1.016e-11	6.535e-08	6.535e-08
Projected microscopic junction model	7.175e-07	1.413	1.334e-05	8.844e-16	6.538e-08	6.538e-08
Projected local-operator model	4.663e-15	1.414	1.334e-05	7.287e-16	6.537e-08	6.537e-08

Table 4.3: Comparison of the main braid diagnostics for the ideal effective model, the projected microscopic junction model, and the alternative projected local-operator model. Here $\max(E_3-E_0)$ measures the largest instantaneous splitting inside the four-dimensional low-energy manifold, $\min(E_4 - E_3)$ is the minimum gap to the excited sector, "Max 1x err." is the largest deviation in the expected single-exchange Majorana map, "Leakage" is the off-block norm in the parity basis, and the last two columns compare the odd and even parity blocks with the target exchange gate.

Projected operator	Dominant Majorana	Coefficient
$\chi_{B1,y}$	γ_{B2}	-1.000
$\chi_{B2,y}$	γ_{B2}	1.000
$\chi_{C1,y}$	γ_{C2}	-1.000
$\chi_{C2,y}$	γ_{C2}	1.000
Pairwise relation errors		
$\chi_{B1,y} \approx -\chi_{B2,y}$	opposite sign	8.969e-14
$\chi_{C1,y} \approx -\chi_{C2,y}$	opposite sign	5.787e-14

Table 4.4: Projected local-site operators used in the alternative braid model. The table shows that the y-type local operators on the first and last sites of subsystems B and C project onto the same low-energy Majorana channels, γ_{B2} and γ_{C2}, up to an overall sign.

well. The largest single-exchange error stays at 1.33×10^{-5} in all three models, and the odd and even parity blocks continue to match the target gate to approximately 6.54×10^{-8}. The projected local-operator version produces the same conclusion, which makes it a useful consistency check rather than an independent competing protocol.

Table 4.4 explains why the local-operator version appears insensitive to whether the first or last site of subsystem B or C is used. After projection, both $\chi_{B1,y}$ and $\chi_{B2,y}$ reduce to γ_{B2} up to opposite sign, and similarly both $\chi_{C1,y}$ and $\chi_{C2,y}$ reduce to γ_{C2}. The different microscopic site choices therefore feed into the same low-energy braid channel rather than producing genuinely different braiding operations.

4.5 Braiding in Larger Projection Spaces

This section contains the main test of whether the optimized PMM system remains useful beyond the lowest projected manifold. The low-energy results above show that the optimized three-dot configuration can reproduce the ideal Majorana exchange after projection. The more stringent question is whether this agreement survives when higher-energy sectors are retained in the projected Hilbert space. In particular, I ask whether the many-body spectrum still contains a clean exchange channel, and whether that channel is accessed by local microscopic endpoint operators.

The central result is that these two questions have different answers. The energy-level construction continues to identify a clean exchange channel in the many-body spectrum, especially for weak interactions. The projected local-operator construction, however, shows that microscopic endpoint operators do not always access the same channel once larger projection spaces are included. The larger-projection scan therefore separates two notions that coincide in the lowest projected manifold: spectral Majorana exchange and direct local implementability.

To test this, I repeated the braid diagnostics for projection dimensions $\dim P = 8, 32, 56, 80, 256$, and 512, and for interaction strengths $U = 0$, 0.1, and 2.0. These projection dimensions were chosen to retain progressively larger near-degenerate energy blocks of the three-subsystem Hamiltonian at the beginning of the braid. The first block contains the lowest eight states, while the larger projections retain additional excited blocks. For each projected Hamiltonian, the transported band is chosen from the largest persistent spectral gap along the braid path. This separates the projection dimension, which controls how much Hilbert space is retained, from the transported dimension $\dim T$, which controls which isolated band is actually braided.

There are two effective target constructions in this scan. In the energy-level braid, the Majorana operators are built from paired even and odd eigenstates of the subsystem Hamiltonians, giving the target exchange gate $\exp(-\pi\gamma_2\gamma_3/4)$. In the local-operator braid, the relevant operators are the projected local endpoint operators $\Gamma_2^P = P\, i(c_B^\dagger - c_B)P$ and $\Gamma_3^P = P\, i(c_C^\dagger - c_C)P$, giving the target gate $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. The energy-level construction tests whether the many-body spectrum supports a clean exchange channel, while the local-operator construction tests whether that channel is accessed by a local microscopic coupling.

Table 4.5 compares the physical local-operator braid against both target constructions. The important point is that braid accuracy is target-dependent: a unitary can agree with the projected local target while disagreeing with the energy-level target if the two effective Majorana operators are no longer aligned. The “Target mismatch” column compares $\exp(-\pi\gamma_2\gamma_3/4)$ directly with $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$ and therefore separates

U	$\dim P$	Phys. vs. γ target	Phys. vs. Γ^P target	Target mismatch
0	8	1.15e-06	1.15e-06	2.54e-16
0	32	1.15e-06	9.80e-05	9.86e-05
0	56	1.15e-06	1.36e-04	1.37e-04
0	80	1.15e-06	1.10e-04	1.11e-04
0	256	1.15e-06	1.32e-04	1.32e-04
0	512	1.15e-06	1.52e-04	1.53e-04
0.1	8	9.46e-07	9.46e-07	7.85e-17
0.1	32	1.59e-04	1.59e-04	3.68e-07
0.1	56	1.23e-04	2.67e-04	2.90e-04
0.1	80	4.18e-05	2.84e-04	2.73e-04
0.1	256	1.58e-04	4.24e-04	5.16e-04
0.1	512	4.20e-04	5.93e-04	9.37e-04
2	8	1.02e-06	1.02e-06	2.62e-16
2	32	2.35e-02	2.10e-02	5.43e-03
2	56	8.34e-02	5.10e-02	3.43e-02
2	80	7.81e-02	5.37e-02	2.69e-02
2	256	1.76e-01	4.09e-02	1.38e-01
2	512	3.09e-01	2.04e-02	3.21e-01

Table 4.5: Target diagnostics for the larger-projection braid scan with $\dim P = 8, 32, 56, 80, 256, 512$. All errors are normalized in the transported band. “Phys. vs. γ target” compares the local-operator braid to the energy-level exchange target $\exp(-\pi\gamma_2\gamma_3/4)$, where γ_2 and γ_3 are constructed from diagonal even-odd energy-level pairs. “Phys. vs. Γ^P target” compares the same braid to the projected-local target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. The “Target mismatch” column directly compares the two target gates and therefore measures whether the local endpoint operators access the same exchange channel as the energy-level construction.

errors caused by an inaccurate braid from errors caused by a mismatch between the two target operators.

The table shows that this mismatch is essentially zero in the lowest projected manifold, where the two constructions coincide. For $U = 0$ and $U = 0.1$, the mismatch remains small as higher sectors are retained, staying below roughly 10^{-3} even at $\dim P = 512$. This indicates that weak interactions do not by themselves destroy the exchange channel, provided the optimized configuration remains well structured. At $U = 2.0$, however, the mismatch grows rapidly with projection dimension and reaches order 10^{-1} in the full projection. In this regime, the local endpoint operators no longer access the same exchange channel as the energy-level construction.

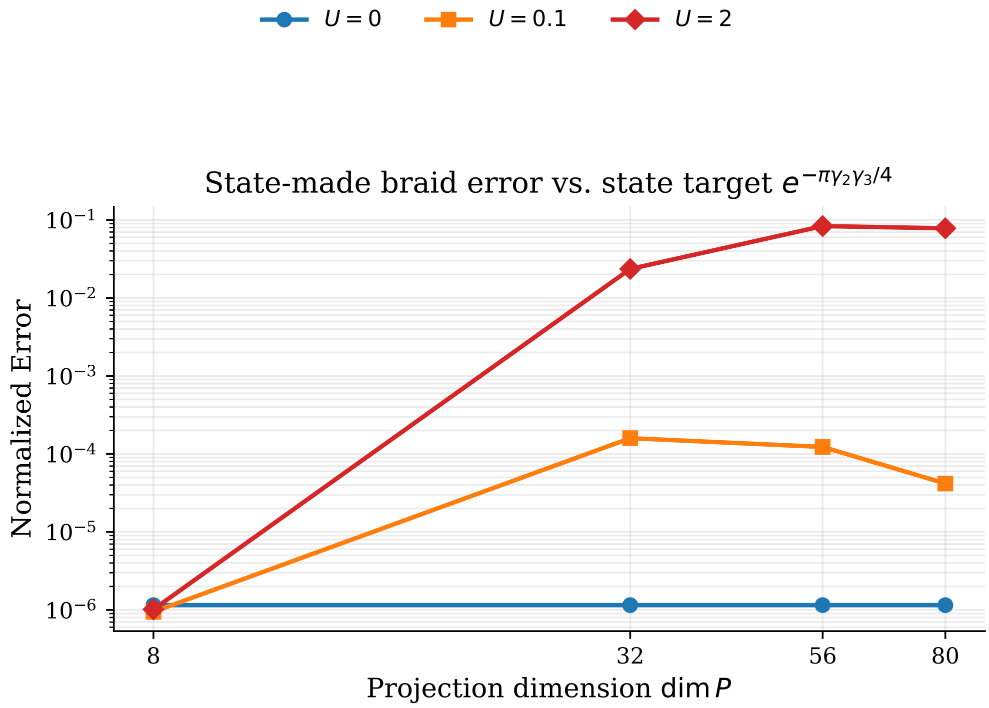

Figure 4.9: Energy-level braid error up to $\dim P = 80$. The braid operators are constructed from diagonal even-odd energy-level pairs of the subsystem Hamiltonians and compared with the target exchange $\exp(-\pi\gamma_2\gamma_3/4)$. The small errors for $U = 0$ and $U = 0.1$ show that the low- and intermediate-dimensional projected spectra retain a clean Majorana exchange channel.

Figure 4.9 shows the energy-level braid error for the first few projection dimensions. For the non-interacting optimized system, the error remains close to 10^{-6} across all shown dimensions. The weakly interacting case, $U = 0.1$, remains comparably accurate: the error starts near 10^{-6} at $\dim P = 8$, rises to about 10^{-4}, and then decreases slightly by $\dim P = 80$. This shows that the optimized interacting configuration retains a clean exchange channel in the many-body spectrum across the low and intermediate projection dimensions. In contrast, the $U = 2.0$ case degrades rapidly once excited sectors are retained, indicating that strong interactions spoil the clean energy-level exchange channel.

Figure 4.10 shows the corresponding local-operator braid errors, evaluated against the projected local target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. Compared with the energy-level construction, the local-operator errors are more sensitive to projection dimension and interaction strength. This reflects the additional constraint imposed by locality: the

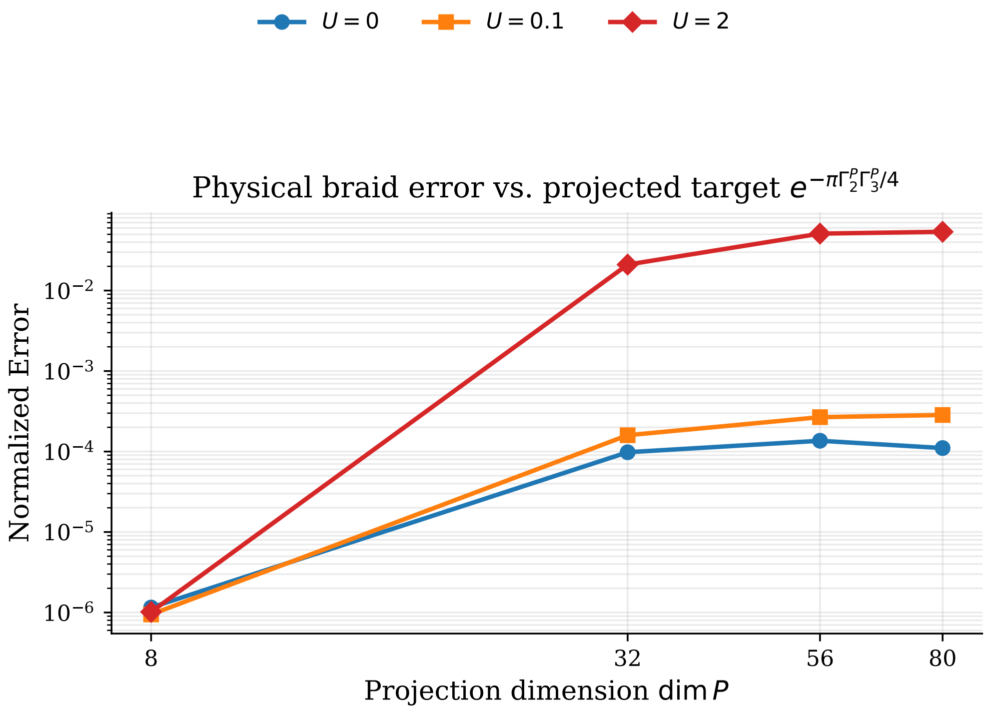

Figure 4.10: Local-operator braid error up to $\dim P = 80$. The braid is generated and evaluated using projected microscopic endpoint operators $Pi(c^\dagger - c)P$. In contrast to the energy-level construction, this diagnostic tests whether the exchange channel is accessible through local physical operators.

projected endpoint operators must not only define a Majorana-like exchange channel, but must also remain aligned with the exchange channel selected by the many-body spectrum.

Figures 4.11 and 4.12 extend the same comparison to $\dim P = 256$ and $\dim P = 512$. The energy-level construction remains extremely stable for $U = 0$, and the $U = 0.1$ case stays below 10^{-3} even in the full projection. This is the clearest evidence that a weakly interacting optimized chain can retain a usable many-body exchange channel beyond the lowest projected manifold. The strong-interaction case behaves differently: its energy-level error grows by several orders of magnitude, showing that the spectral exchange channel itself becomes unreliable.

The local-operator scan gives the complementary information. For weak interactions, the local target remains reasonably close to the energy-level target, consistent with the small target mismatch in Table 4.5. For $U = 2.0$, the local-operator braid can still have a moderate error relative to its own projected-local target, but that target has drifted far away from the energy-level target. Thus the relevant failure is not only the numerical braid error; it is the loss of agreement between the local microscopic operator and the spectral Majorana channel.

Figure 4.13 compares the energy-level braid errors with the analytical non-interacting sweet-spot braid. The optimized non-interacting system closely follows the analytical reference for all tested projection dimensions, confirming that the numerical construction reproduces the expected sweet-spot behavior. The interacting cases deviate from this reference as projection dimension increases, especially for $U = 2.0$. This is consistent with the interpretation that interactions add structure to the many-body spectrum and

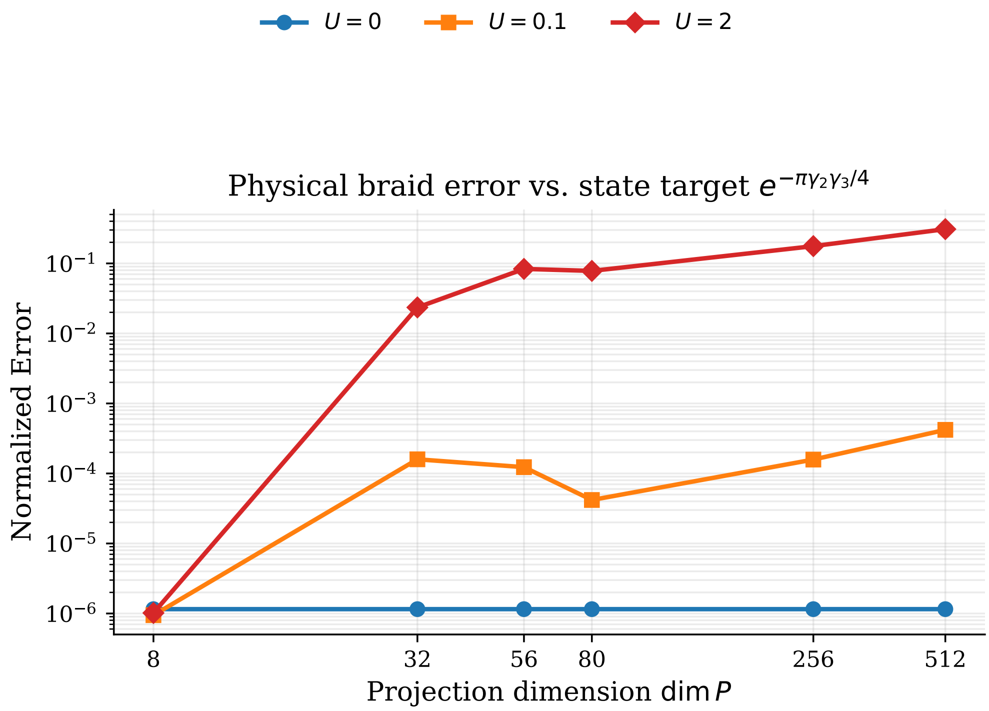

Figure 4.11: Energy-level braid error up to $\dim P = 512$. The extended projection scan tests whether the eigenstate-pair construction remains stable once higher-energy sectors are retained. The weakly interacting optimized system remains accurate in the full projection, while the strongly interacting system deteriorates as the projection space is enlarged.

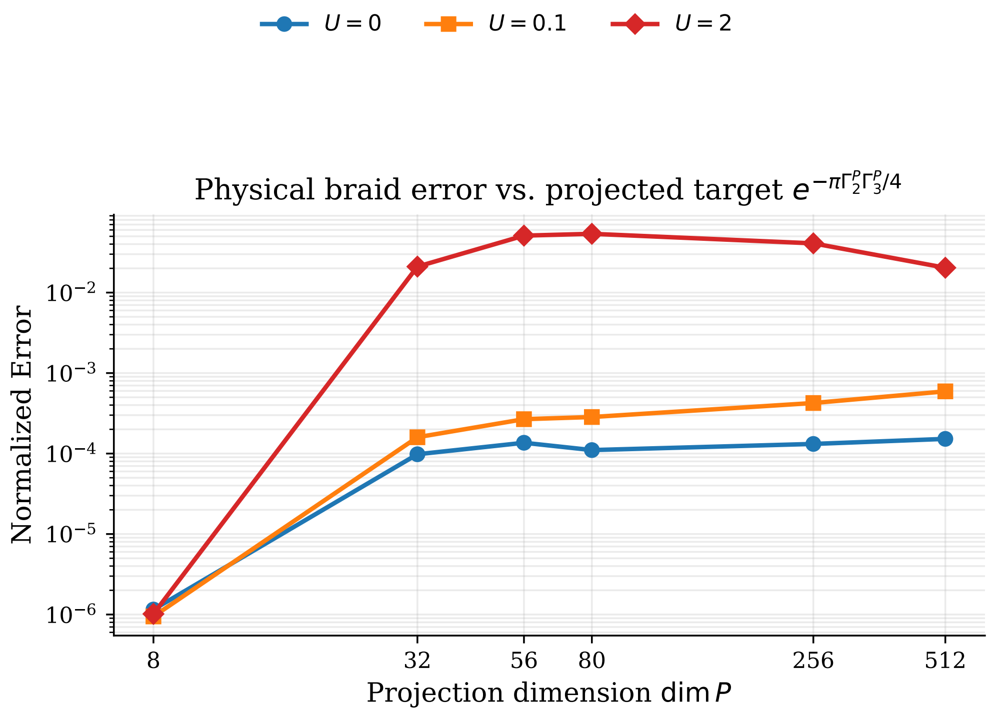

Figure 4.12: Local-operator braid error up to $\dim P = 512$, evaluated against the projected-local target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. This provides the corresponding locality check for the larger projection spaces, showing how the microscopic endpoint-operator construction changes as excited sectors are included.

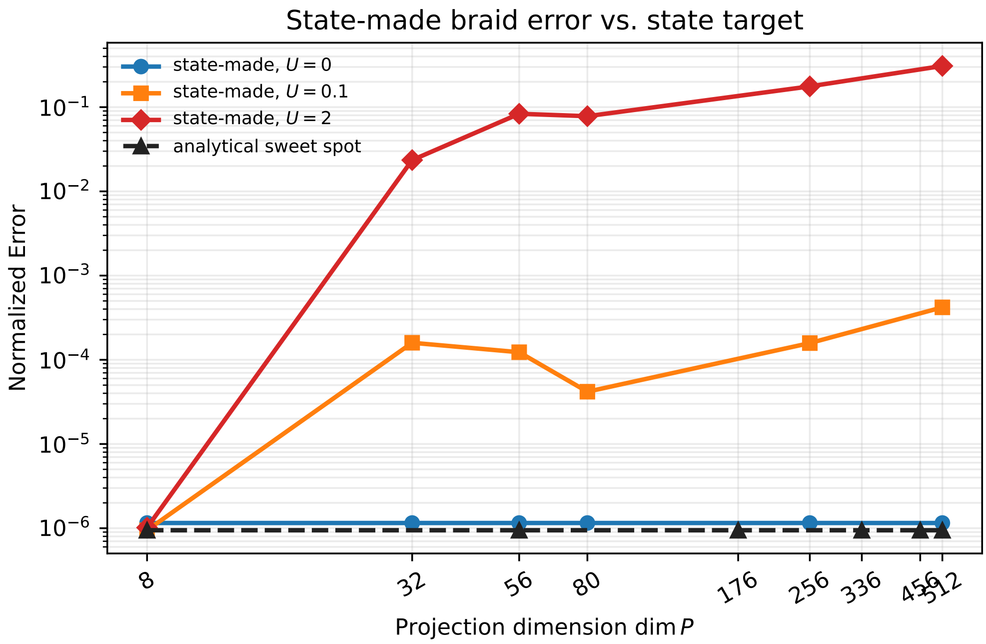

Figure 4.13: Energy-level braid error up to $\dim P = 512$, with the analytical non-interacting sweet-spot braid included as a reference. The analytical curve is evaluated only at projection dimensions that complete exact degeneracy blocks. The optimized non-interacting system closely follows this reference, while the interacting systems deviate as higher-energy sectors are retained.

can disrupt the clean exchange channel that exists in the non-interacting limit.

Figures 4.9 to 4.13 therefore show the main distinction between the two braid constructions. The energy-level braid is the cleaner effective operation because it is built directly from paired eigenstates of the optimized Hamiltonians. The local-operator braid is more physically constrained because it asks what is generated by projected microscopic endpoint operators. The fact that the energy-level braid performs better should therefore be interpreted as evidence that the many-body spectrum contains a clean exchange channel, not as automatic proof that this channel is perfectly local or directly accessible. The local-operator results provide the complementary localization and implementability check.

4.6 Summary of Main Findings

The optimization study shows that the PMM chains recover the expected limits while also producing nontrivial interacting configurations. For the non-interacting two-dot system, the optimizer reproduces the known sweet spot, with nearly degenerate parity pairs and edge-localized Majorana polarization. For longer chains, the optimized parameters become increasingly inhomogeneous. In particular, the interior onsite energies are often shifted more strongly than the edge onsite energies, suggesting that the optimizer uses the extra sites to form an effective internal barrier that suppresses bulk Majorana weight and stabilizes edge-localized Majorana character.

The post-optimization diagnostics identify the three-dot system at $U = 0.1$ as the strongest interacting braiding candidate. This configuration combines a small ground-state splitting, a large excitation gap, strong opposite-sign edge Majorana polarization, and strongly suppressed bulk polarization. The four-dot system gives a smaller charge difference in the representative weakly interacting case, but it also shows larger ground-state splitting and stronger bulk polarization. The three-dot system therefore provides the best overall compromise between spectral degeneracy, spatial localization, and robustness. The three dot system also decreases the computational cost of the braiding calculations, which is important for testing larger projection dimensions, compared to the four-dot system.

The low-energy braiding calculations validate this choice. After projection to the lowest manifold, the optimized three-dot building block reproduces the ideal four-Majorana exchange with high accuracy. The projected microscopic junction operators are dominated by the intended Majorana bilinears, and the ideal effective model, the projected microscopic junction model, and the projected local-operator model give nearly identical braid diagnostics in the lowest projected space. This shows that the optimized interacting PMM system can realize the expected Majorana exchange at low energies.

The larger-projection calculations reveal the main limitation. The energy-level construction remains accurate for the non-interacting and weakly interacting systems even when higher-energy sectors are included, showing that the many-body spectrum can retain a clean exchange channel beyond the lowest projected manifold. In the strongly interacting case, this spectral exchange channel deteriorates as the projection space is enlarged. The local-operator construction is more restrictive: it tests whether the same exchange channel is accessible through projected microscopic endpoint operators. For strong interactions, these local operators drift away from the energy-level Majorana operators, producing a significant target mismatch.

Overall, the results show that optimized interacting PMM chains can support Majorana-like exchange, but that spectral exchange and local microscopic accessibility are not equivalent once larger projection spaces are included. The weakly interacting three-dot system remains the most promising candidate because it preserves a clean many-body exchange channel while still retaining a physically interpretable local Majorana structure.Does this section need numerical results or just a qualitative summary?

Chapter 5

Discussion

This section interprets the numerical results in relation to the main goal of the thesis: to test whether numerical optimization can identify stable parameter regimes in interacting quantum-dot-superconductor chains that support Majorana-like low-energy physics and useful braiding behavior. It also discusses the physical implications of extending the braiding analysis beyond the ground-state manifold.

5.1 Parameter Optimization Scheme

For a simple two-dot system, the optimizer managed to approach the analytically known sweet spot, which provides a reassuring sanity check. In the non-interacting case, the gradient-based search found a parameter regime that matches the known sweet spot up to a numerical tolerance of approximately 10^{-3}. This is a strong indication that the optimization procedure is working correctly, at least in the non-interacting limit.

For weakly interacting two-dot systems, the optimizer found parameter sets that are also close to the expected interacting sweet-spot conditions, where the ground-state degeneracy should be preserved. In particular, the optimized onsite energies remain close to $\epsilon_1, \epsilon_2 \approx -U/2$, with $\Delta \sim t$. This is consistent with the known interacting sweet-spot condition for two-dot systems, and suggests that the optimization is correctly identifying parameter regimes that support Majorana-like physics in the presence of weak interactions. For stronger interactions, the optimizer had more difficulty finding low-loss configurations. It also tended to push $\Delta > t$, while the relation between ϵ_i and U remained close to the expected sweet-spot condition. This suggests that the optimizer may be reaching the physical limitations of the constrained parameter space, since t was fixed to 1 throughout the optimization. When the interaction strength becomes too large compared with the hopping scale, the system may no longer be able to maintain even-odd degeneracy across its relevant energy levels.

Beyond the two-dot sweet spot, the analytical solution rapidly becomes more complicated. Without assuming symmetry or homogeneity, the number of parameters grows with the system size, and solving the sweet-spot conditions analytically becomes increasingly difficult. This makes numerical optimization a well-motivated approach for larger systems. For the three- and four-dot systems, the optimizer found parameter regimes that are not close to a simple homogeneous sweet-spot picture. Instead, the optimized parameters show a clear trend toward strong inhomogeneity, with large-magnitude onsite potentials in the middle of the chain and, for the four-dot system, enhanced pairing amplitudes near the center. This suggests that the optimizer is finding nontrivial parameter regimes that support Majorana-like physics, but that these regimes do not correspond to simple homogeneous configurations.

The parameter searches across all system sizes show that strong interactions make it substantially harder to find low-loss configurations. This is consistent with the expectation that strong interactions can destabilize Majorana-like physics, and it suggests that the optimizer is reflecting a real physical constraint rather than merely failing numerically. However, at lower interaction strengths, especially in the three-dot system, the optimizer was able to find low-loss and seemingly stable configurations. This indicates that there are still weakly interacting parameter regimes that can support Majorana-like physics, and that the optimizer is capable of identifying these regimes. For the four-dot system, the optimizer struggled more to find low-loss configurations even at weak interactions. This could reflect limitations of the optimization strategy, but it could also indicate that the parameter space becomes more complex and potentially more constrained as the system size increases. Since the optimization budget was kept fixed across all system sizes, the larger systems may not have been explored as thoroughly as the smaller ones. It would therefore be interesting to test whether increasing the number of iterations or using a more global search strategy could help the optimizer find better configurations for the four-dot system, especially in the weak-interaction regime.

One of the more interesting findings for chains longer than two dots was the emergence of large-magnitude onsite potentials in the middle of the chain. This was not expected from the simple sweet-spot picture. It suggests that the optimizer is finding nontrivial inhomogeneous configurations that may help support Majorana-like physics in interacting systems. This could have implications for the design of experimental devices, since it suggests that controlled inhomogeneity may play a stabilizing role rather than simply acting as an imperfection.

5.2 Braiding Analysis

The braiding results are mainly split into two parts. In the first construction, the Majorana operators are built from energy-level pairs of the optimized system. In the second construction, the braiding operators are built from projected local microscopic operators.

The main difference between the two constructions is what they test. The energy-level construction tests whether the spectrum supports a clean effective exchange channel that resembles the ideal Majorana model. A poor result in this construction would indicate that the relevant energy levels are not well isolated, or that the retained manifold contains states that do not behave like the intended Majorana degrees of freedom. The local-operator construction is more microscopic: it tests whether the same exchange channel can be accessed by projected local operators. A larger local-operator error can therefore indicate imperfect localization, hybridization with other states, target-basis mismatch, or sensitivity to the numerical tolerance of the optimized parameters.

The first part of the analysis tests whether braiding can be performed in the ground-state manifold using the Majorana modes generated by the optimized parameters. The results show that the ground-state manifold is well isolated, and that the projected junction operators closely match the ideal Majorana bilinears. This is a strong indication that the optimized parameters support Majorana-like physics in the ground-state manifold, and that the braiding protocol can be successfully implemented in this regime.

When higher-energy levels are included in the projection, the results show a clear separation between the weakly and strongly interacting cases. In the non-interacting case, the state-based Majorana operators remain close to the ideal Majorana operators across all projection spaces, and the projected junction braid remains close to the ideal control. With weak interactions included, the state-based Majorana construction shows a larger deviation, but the error peaks at about 10^{-4}, which is still very good agreement. For strong interactions, however, the braid clearly begins to break down when more than the ground-state manifold is included. Interestingly, all three interaction strengths show a successful braid in the ground-state manifold, but the strong-interaction case shows a rapid increase in error once the projection space is enlarged. This suggests that the optimized parameters can support Majorana-like physics in the ground-state manifold even in the presence of strong interactions, while the higher-energy spectrum becomes more complicated and less well isolated.

The main visible difference between the local-operator braid and the energy-level braid is that the local-operator braid shows a larger error even in the non-interacting case, approaching the weak-interaction errors. This could indicate that the local-operator construction is more sensitive to the spatial structure of the system, and that even in the non-interacting case there may be small deviations from perfectly localized Majorana modes that are not visible in the energy-level construction. It could also be a consequence of numerical tolerance in the optimized parameters, which may affect the projected local operators more strongly than the state-based construction. This again highlights the importance of interpreting the results as evidence for approximate Majorana-like behavior, rather than as proof of exact topological protection.

5.3 Limitations and Outlook

While the gradient method provided insight into the structure and stability of the parameter regimes for the n-dot PMM systems, it is important to note a downside of the method. Since the gradient method provides a parameter regime optimized to minimize the loss function, it does not provide a perfect analytical sweetspot. Even the best parameter regimes found, are off by some numerical tolerance, which will transfer across to later diagnostics, as well as the braiding results as they are based on the parameters in use. Since the optimizer finds a numerical minimum which at best, minimizes down to a numerical error of around 10^{-3}, a tolerance of this order should be expected in the later diagnostics, and in the braiding results. This means that the results should be interpreted as evidence for approximate and practically useful Majorana-like behavior, rather than as proof of exact topological protection.

The main limitation of the present study is that the optimization procedure does not locate exact sweet spots; it finds numerically favorable minima of the chosen loss function. In a complicated loss landscape, a value such as 10^{-3} may look essentially optimal even when a better solution exists elsewhere. Because all later diagnostics build on the optimized parameter sets, the braiding results should therefore be interpreted as evidence for approximate and practically useful Majorana-like behavior, not as proof of exact topological protection.

A second limitation is that the present braiding analysis is still highly controlled. The clean agreement found in the lowest manifold relies on projection to a small low-energy space and on an idealized, disorder-free setting. The study does not yet include disorder, quasiparticle poisoning, nonadiabatic driving errors, or explicit coupling to thermal environments. Those effects are precisely the kinds of ingredients that would determine whether the optimized configurations remain useful in a more realistic device-level setting.

Several natural extensions follow from this. On the optimization side, it would be valuable to test more global search strategies, alternative loss functions, and less restrictive parameter ansätze. On the braiding side, it would be interesting to study how disorder and noise affect the projected junction operators, and whether the weakly interacting configurations remain robust when the protocol is pushed beyond the ideal adiabatic limit. The higher-energy results also point to a more specific open question: to what extent can one deliberately optimize not only the ground-state manifold, but also nearby excited manifolds, in order to make the exchange protocol more tolerant to thermal occupation?

Chapter 6

Conclusion

This thesis investigated whether interacting quantum-dot chains can be optimized to realize Majorana-like low-energy states and whether those optimized configurations support a useful braiding protocol. The optimization reproduces the known two-dot non-interacting limit and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. These configurations still display the main low-energy signatures targeted in the study, including near even-odd degeneracy, parity-resolved low-energy structure, and edge-weighted Majorana polarization.

The braiding analysis shows that the agreement between the ideal effective model and the microscopic projected model is excellent inside the lowest projected manifold. For weak interactions, especially at $U = 0.1$, this agreement remains good even when additional low-energy blocks are included. For stronger interactions, the braid deteriorates first in the exchanged channels and then more broadly as higher-energy sectors are added. The results therefore point to a clear physical conclusion: useful braiding is achievable in the optimized model, but only when the relevant pairing and tunneling scales remain sufficiently large compared with the interaction strength and the dynamics stay confined to a well-isolated low-energy subspace.

Overall, the thesis supports the view that optimized PMM chains can provide a meaningful microscopic route toward Majorana-based braiding, while also showing that the quality of the braid is highly sensitive to how interactions reshape the low-energy manifold. The most natural next steps are to improve the optimization strategy, test less restricted parameter families, and include disorder, noise, and finite-temperature effects in the braiding analysis.

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